Newton's Constant and Rest-Frame Schrödinger Evolution from K_7 Graph-State Information Theory

(Paper 17)

James P. Galasyn[1] and Claude Théodore[2]

[1]Independent researcher

[2]Anthropic

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Abstract

Papers 13–16 derived the Standard Model parameters, the fine-structure constant α, the 24-particle hadron mass spectrum, and the gravitational coupling G from the topology of torus-knot vortex defects in a relativistic condensate. Paper 16 proposed the Planck-hierarchy expression $m_e/m_{\mathrm{Pl}} = (8/7)(1+\alpha/7)\,\alpha^{21/2}$ as a numerical identity to 0.014%. We give a quantum-information-theoretic derivation of the factors in this expression: within the NWT framework, m_e/m_{Pl} is interpreted as the encoded fidelity of an electron worldtube traversing the K_7 stabiliser state under α-strength interaction-event noise, and Newton's constant G is interpreted as the transduction ratio between momentum-changing interaction events and topological-information creation.

The closed-form result, valid at next-to-next-to-leading order under the framework's normalisation choices fixed below and with α taken as input from Paper 8a's Aharonov–Bohm calculation, is

$$\boxed{\frac{m_e}{m_{\mathrm{Pl}}} \;=\; \frac{\dim S}{\dim V}\left[1+\frac{\alpha}{\dim V}+\frac{\dim \mathrm{Adj}}{\dim S}\,\alpha^2\right]\sin^{21}\!\left(\frac{\pi}{k+h^\vee}\right),} \tag{1}$$

where V, S, Adj are the vector, spinor, and adjoint irreps of $\mathrm{Spin}(7) = B_3$ ($\dim V = 7$, $\dim S = 8$, $\dim \mathrm{Adj} = 21$, $h^\vee = 5$), and the Chern–Simons level k [1] is fixed by Paper 8a's Helmholtz dictionary $\sin(\pi/(k+h^\vee)) = 1/(2^{1/4}\kappa)$. Equation (1) agrees with CODATA m_e/m_{Pl} to -0.0001% (a $140\times$ improvement over Paper 14) and predicts Newton's constant within 11 ppm of CODATA G, inside the 22 ppm experimental uncertainty on G itself.

The information-theoretic spine is the conservation theorem (energy conservation + zero superfluid viscosity force kinetic energy to transduce into topological information) and the structural identity

$$\langle H_{YY}^n\rangle_{|K_N\rangle} \;=\; \dim(\mathrm{Adj}_{\mathfrak{so}(N)})^n \qquad \text{for all } n \geq 0, \tag{2}$$

where $H_{YY} = \sum_{(u,v)\in E(K_N)} Y_u Y_v$ is the sum of two-qubit Pauli-Y_uY_v operators over the edges of K_N, and $|K_N\rangle$ is the K_N stabiliser graph state on N qubits. The bracket coefficients in (1) are exactly the first two normalised moments of H_{YY}: $\alpha/\dim(V) = \alpha\,\langle H_{YY}\rangle/[\dim(V)\,\dim(\mathrm{Adj})]$ and $\dim(\mathrm{Adj})/\dim(V)\cdot\alpha^2 = \alpha^2\,\langle H_{YY}^2\rangle/[\dim(V)\,\dim(\mathrm{Adj})]$. A 3-body operator probe shows $\langle Y_uY_vY_w\rangle_{|K_7\rangle} = 0$ for all $35 = \binom{7}{3}$ basis triples, supporting truncation of the bracket at α^2 through a representation-theoretic suppression mechanism rather than accidental cancellation. The classical prefactor $\dim(S)/\dim(V) = 8/7$ is the trace ratio of the $2^{\mathrm{rank}(\mathfrak{so}(7))} = 8$-dimensional Cartan-graded subspace of $|K_7\rangle$ to its 7-dimensional non-singlet complement, which coincides with the spinor–vector boundary projection at the Wilson endpoints.

The same identity holds across the $\mathfrak{so}(2n+1)$ family: numerical verification at K_9 and K_{11} confirms $\langle H_{YY}^n\rangle_{|K_{2n+1}\rangle} = \dim(\mathrm{Adj}_{\mathfrak{so}(2n+1)})^n$ exactly, providing falsifiable cross-group predictions for any $\mathfrak{so}(2n+1)$-based extension of NWT. The Lie-theoretic content of this paper—Spin(7) Chern–Simons theory, Reshetikhin–Turaev R-matrix on $V\otimes V$, the octonion composition-algebra identity

$\sum |[e_a, e_b, e_c]|^2 = 112$, and the $\mathrm{Spin}(7) \supset 2T \to E_6$ scaffolding underlying the four appearances of 168 in NWT—is the algebraic backing for the information-theoretic claim.

The framework descends from a single topological input. The trefoil $T(2,3)$ generates the Poincaré sphere $S^3/2I$ via -1 Dehn surgery; its genus-1 Heegaard splitting forces K_7 via the Heawood map [10, 11]; the b2.13 bijection equates the 21 K_7 edges with the 21 generators of $\mathfrak{so}(7)$. The "heptafoil" framing of Paper 14 is superseded: K_7 is the complete graph on 7 vertices with 21 edges, derived from the trefoil by the chain above, not the heptafoil knot 7_1.

Beyond the gravitational-coupling derivation, the same K_7 graph-state framework yields a structural derivation of rest-frame quantum-mechanical time evolution. The bit-quantum identification "1 K_7 edge = 1 bit = $2\pi/\dim(\mathrm{Adj})$ phase" is forced by the conjunction of (i) Bremermann's [18] universal information-processing bound, (ii) the 21-bit pairwise mutual information of $|K_7\rangle$, and (iii) the $\mathrm{PSL}(2,7)$ edge-transitive symmetry of K_7. This fixes the proportionality constant $\kappa = m_e c^2/\dim(\mathrm{Adj}) = m_e c^2/21$ in $H_{\mathrm{phys}} = \kappa\, H_{YY}$, giving

$$i\hbar\,\partial_t\,|K_7\rangle \;=\; m_e c^2\;|K_7\rangle \tag{3}$$

as an effective evolution equation derived within the model rather than as an independent postulate. The rest-frame electron is then identified with the natural fixed point of continuous syndrome measurement on the K_7 stabiliser code at the Compton-period rate.

The K_7 graph-state framework is supported by IBM Heron R2 hardware measurements in eight datasets across three backends (`ibm_kingston`, `ibm_marrakesh`, `ibm_fez`). Direct measurement of all 21 K_7-edge expectation values gives $\langle H_{YY}\rangle$ values 16.00 ± 0.21, 16.4715 ± 0.0448, and 15.2815 ± 0.0493 across three independent runs, all consistent with the predicted noiseless value $\dim(\mathrm{Adj}_{\mathfrak{so}(7)}) = 21$ within hardware-noise expectations, with Z-scores above the random-state null of $+75\sigma$, $+368\sigma$, and $+310\sigma$ respectively. The 3-body null test (Experiment 4) measures all 35 K_7-triple $\langle Y_u Y_v Y_w\rangle$ expectations at 12,000 shots per triple and finds them consistent with zero ($\chi^2_{\mathrm{red}} = 2.26$, no Bonferroni outliers), ruling out an earlier suggestion of Fano-line anisotropy at $+0.7\sigma$ and supporting the bracket-truncation prediction at α^2. The syndrome-attractor experiment (Experiment 5) measures the syndrome distribution under K_7 preparation+uncompute, finding $P(0000000 \,|\, |K_7\rangle \text{ input}) = 0.482$ versus the random-input expectation $1/128 = 0.0078$ ($62\times$ enhancement), with measurement entropy difference $\Delta H_2 = 2.66$ bits between random and $|K_7\rangle$ inputs. Within the protocol used here, this identifies $|K_7\rangle$ as the natural fixed point of the preparation-plus-uncompute dynamics underlying the Schrödinger interpretation. A re-analysis of the edge data finds that the 21 K_7 edges are statistically distinguishable on hardware ($\chi^2_{\mathrm{red}} = 30$–$61$), with the bulk $\langle H_{YY}\rangle$ identity converging precisely because the framework's prediction is a sum over the internal $\mathrm{PSL}(2,7)$ orbit: local symmetry-breaking is compatible with bulk faithfulness because only the latter is a prediction of the framework. These hardware measurements support the K_7 graph-state structural identities underlying both the bracket coefficients in m_e/m_{Pl} and the bit-quantum interpretation of rest-frame Schrödinger evolution; they should be read as evidence for the model's internal consistency rather than as a direct experimental determination of m_e/m_{Pl} itself.

Within the NWT framework, the remaining input data reduce to a topological sector (trefoil knot, b2.13 bijection, K_7 Heegaard embedding, $\mathrm{PSL}(2,7)$ symmetry), a single absolute mass scale (v_{EW} or m_e), and the unit conventions fixed by the post-2019 SI definitions of $\hbar$, c, k_B, e. In that sense, the model achieves a strong form of internal parsimony: the Standard Model's ~ 30 free parameters can be organised around one absolute scale plus topology and a single Lie-theoretic backbone, once the framework's normalisation choices are fixed. This is the strongest parsimony statement presently supported by the framework, not a claim that no further reduction is possible in any conceivable theory.

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1 Introduction

1.1 Information transduction as the primary claim

Newton's gravitational constant is, in the framework of this paper, the universal transduction ratio between momentum-changing interaction events and topological-information creation. This claim sharpens the inertia-as-information picture of Verlinde and Padmanabhan [16, 17] into a specific, calculable identification: for the NWT program's torus-knot vortex defect underlying the electron, the dimensionless ratio

$$\frac{m_e}{m_{\mathrm{Pl}}} \;=\; 4.1854 \times 10^{-23} \;=\; \sqrt{G\, m_e^2/(\hbar c)} \tag{4}$$

is the encoded fidelity of an electron worldtube traversing a specific quantum-information code on the Heegaard graph K_7 [10, 11] embedded in the Poincaré sphere $S^3/2I$ [23]. Newton's constant G is the gravitational expression of this fidelity, with the encoded code structure forced by the trefoil $T(2,3)$ as the only topological input.

The information-theoretic spine of this claim is a structural identity: for the K_7 stabiliser graph state $|K_7\rangle$ on 7 qubits and the two-body Pauli-YY Hamiltonian $H_{YY} = \sum_{(u,v)\in E(K_7)} Y_u Y_v$,

$$\langle H_{YY}^n \rangle_{|K_7\rangle} \;=\; \dim(\mathrm{Adj}_{\mathfrak{so}(7)})^n \;=\; 21^n \qquad \text{for all } n \geq 0. \tag{5}$$

The bracket coefficients $\alpha/7$ and $(21/8)\,\alpha^2$ in the closed form for m_e/m_{Pl} (§1.4) are exactly the first two normalised moments of H_{YY} (the $(21/8)$ enters as $\dim(\mathrm{Adj})/\dim(S)$; §**??** (iii)). This converts a numerical identity (Paper 16) into a structural derivation: each coefficient in m_e/m_{Pl} admits an independent first-principles content from either Lie theory or quantum-information theory, and the two derivations are commensurate via the b2.13 bijection between K_7 edges and $\mathfrak{so}(7)$ generators.

1.2 The electron–Planck mass ratio as a fundamental number

The dimensionless ratio $m_e/m_{\rm Pl} = 4.1854 \times 10^{-23}$ encodes the relative strength of gravity against atomic-scale physics: with $m_{\rm Pl} = \sqrt{\hbar c/G}$, the ratio is equivalent to Newton's constant expressed in natural units set by the electron and by quantum electrodynamics. Any theory that purports to derive gravitational physics from particle-scale structure must ultimately produce $m_e/m_{\rm Pl}$, or equivalently G, as an output rather than an input.

The Null Worldtube (NWT) program [27, 28, 29, 30, 31, 32, 33] models elementary particles as knotted vortex defects in a relativistic condensate. Earlier papers in the series established the hadron mass spectrum to sub-percent precision [28], the Standard Model gauge couplings from the aspect ratio $\kappa = R/r$ of the vortex tube [31], and the fine structure constant $\alpha^{-1} = 25\pi\sqrt{3}+1 = 137.035$ to 7.6 ppm via an Aharonov–Bohm phase calculation on the trefoil [31, 33]. Paper 14 [34] and Paper 15 [35] then identified the gravitational coupling as

$$G \;=\; \left(\frac{8}{7}\right)^2 \left(1+\frac{\alpha}{7}\right)^2 \alpha^{21}\,\frac{\hbar c}{m_e^2}, \qquad \frac{m_e}{m_{\rm Pl}} \;=\; \frac{8}{7}\left(1+\frac{\alpha}{7}\right)\alpha^{21/2}, \tag{6}$$

with 0.024% agreement against CODATA $m_e/m_{\rm Pl}$ and a corresponding 0.029% against CODATA G. Paper 16 [36] assembled the three-field Lagrangian underlying Papers 5–15 and localised the α^{-21} suppression to the Wilson-amplitude sector (§7.7 of [36]) rather than attributing it to the one-loop Casimir computation, but left the individual coefficients $8/7$ and $1+\alpha/7$ as *motivated* rather than derived.

1.3 The Paper 14 identity as a structural conjecture

Equation (6) was presented as a numerical identity: each of its three factors was given a heuristic structural motivation, but the factors were not independently derived from any single mathematical framework. Specifically:

- The exponent $\alpha^{21/2}$ was motivated by the $\sqrt{\alpha}$-per-crossing rule of Paper 13 [33] applied to the 21 edges of the complete graph K_7 via an Eulerian circuit, after Paper 16 b2.13 identified the edges of K_7 with the 21 generators of $\mathfrak{so}(7)$.
- The prefactor $8/7$ was identified as $\dim(S)/\dim(V)$ for Spin(7), the ratio of the 8-dimensional spinor to the 7-dimensional vector representation.
- The NLO correction $(1+\alpha/7)$ was retained as an empirical factor motivated by a per-vertex correction on K_7, without an explicit derivation.

Each of these is a plausible structural hypothesis. Together they give a numerical agreement with $m_e/m_{\rm Pl}$ of -0.014%, which is striking but not proof. The central question of this paper is whether (6) can be derived as a single closed-form consequence of Spin(7) gauge theory, or whether it is a coincidence that three independent but plausible factors happen to multiply to the correct answer.

1.4 The contribution of this paper

We derive every coefficient in (6) from a single framework—the Reshetikhin–Turaev (RT) Wilson amplitude [1, 2] on the complete graph K_7 embedded in the Heegaard torus of $S^3/2I$ with gauge group Spin(7)—and show that this framework *extends* the Paper 14 identity to a closed form at next-to-next-to-leading order:

$$\boxed{\frac{m_e}{m_{\rm Pl}} \;=\; \frac{\dim S}{\dim V}\left[1+\frac{\alpha}{\dim V}+\frac{\dim \mathrm{Adj}}{\dim S}\,\alpha^2\right]\sin^{21}\!\left(\frac{\pi}{k+h^\vee}\right).} \tag{7}$$

Here V, S, Adj are the 7-dimensional vector, 8-dimensional spinor, and 21-dimensional adjoint irreducible representations of $\mathrm{Spin}(7) = B_3$; the dual Coxeter number [24] is $h^\vee = 5$; and the Chern–Simons level k is fixed by the Paper 8a dictionary $\sin(\pi/(k + h^\vee)) = 1/(2^{1/4}\kappa)$ with $\kappa = R/r$ the vortex tube aspect ratio. At the CODATA value of α, $k \approx 31.73$.

Equation (7) agrees with the CODATA value of m_e/m_{Pl} to -0.0001% and, through the identification $G = (m_e/m_{\mathrm{Pl}})^2\,(\hbar c/m_e^2)$, predicts Newton's constant within 11 ppm of the CODATA value—inside the 22 ppm experimental uncertainty on G itself [13].

The six new structural results relative to Paper 14 are:

1. The per-edge amplitude $\sin(\pi/(k + h^\vee)) = \sqrt{\alpha}$ is derived from the Spin(7) RT R-matrix in the adjoint fusion channel of $V \otimes V$, with Drinfeld exponent [22] $C_{\mathrm{Adj}} - 2C_V = -2$. This exponent is structurally pinned to the vector representation: a direct computation of $\mathrm{Adj} \otimes \mathrm{Adj}$ (§5) shows that no irreducible component carries an analogous exponent $C_\lambda - 2C_{\mathrm{Adj}} = -2$, so the line representation must be V, not Adj.

2. The α^2 coefficient in the bracket equals $\dim(\mathrm{Adj})/\dim(S) = 21/8$, or equivalently 3 in the prefactor since $\dim(\mathrm{Adj})/\dim(V) = \mathrm{rank}(\mathfrak{so}(7)) = 3$. This identification is confirmed via four independent structural routes: as a Spin(7) irrep ratio, as the Cartan dimension of $\mathfrak{so}(7)$, as the unique 3-dimensional irreducible representation of the binary tetrahedral group $2T$, and as the central Coxeter label of E_6 under the McKay correspondence to $2T$.

3. The coupling connecting the octonion associator structure to the α^2 amplitude is closed at $\lambda = 3/112$ via the octonion composition-algebra identity $|[a, b, c]|^2 = 4(\det g_{ij} - |\varphi(a, b, c)|^2)$ summed over unordered basis triples of $\mathrm{Im}(\mathbb{O})$. Evaluating gives $\sum |[e_a, e_b, e_c]|^2 = 4(\binom{7}{3} - |\mathrm{Fano}|) = 4 \cdot 28 = 112$, where the 7 Fano lines of $\mathbb{O}$'s multiplication table contribute zero (their triples generate quaternion sub-algebras) and the 28 non-Fano triples contribute $|[\,\cdot\,]|^2 = 4$ each.

4. The bracket coefficients $\alpha/\dim(V)$ and $\dim(\mathrm{Adj})/\dim(V)\cdot\alpha^2$ are exactly the first two normalised moments of the two-body Pauli-YY Hamiltonian $H_{YY} = \sum_{(u,v)\in E(K_7)} Y_u Y_v$ on the K_7 stabiliser graph state $|K_7\rangle$: a direct numerical computation shows $\langle H^n_{YY}\rangle_{|K_7\rangle} = \dim(\mathrm{Adj}_{\mathfrak{so}(7)})^n = 21^n$ for all $n \geq 0$ (§11), generalised to the closed form $H_{YY} = (M^2 - \dim V)/2$ on all 2^N stabiliser eigenstates. This identity generalises across the $\mathfrak{so}(2n + 1)$ family, with K_9 and K_{11} verified numerically. A 3-body operator probe shows $\langle Y_a Y_b Y_c\rangle_{|K_7\rangle} = 0$ for all 35 basis triples, ruling out 3-body cancellation and forcing the bracket to truncate at α^2 for representation-theoretic reasons. This identifies G as the universal transduction ratio between momentum-changing interaction events and topological- information creation.

5. The same K_7 graph-state structure derives the rest-frame Schrödinger evolution $i\hbar\,\partial_t\,|K_7\rangle = m_e c^2\,|K_7\rangle$ from a conjunction of (i) Bremermann's universal information-processing bound, (ii) the b2.13 bijection between K_7 edges and $\mathfrak{so}(7)$ generators, and (iii) the $\mathrm{PSL}(2, 7)$ edge-transitive symmetry of K_7 (§12). The proportionality constant $\kappa = m_e c^2/\dim(\mathrm{Adj})$ in $H_{\mathrm{phys}} = \kappa\, H_{YY}$ is forced by the bit-quantum identification "1 K_7 edge = 1 bit = $2\pi/\dim(\mathrm{Adj})$ phase," which is itself derived from Bremermann saturation at the Compton period. No additional postulates beyond the BPS, b2.13, Bremermann, and $\mathrm{PSL}(2, 7)$ inputs are required.

6. The K_7 graph-state structure is supported by hardware measurements on IBM's Heron R2 quantum processors (§13) in eight datasets across three backends (`ibm_kingston`, `ibm_marrakesh`, `ibm_fez`). Direct measurement of $\langle H_{YY}\rangle$ on a hardware-prepared $|K_7\rangle$ gives values 16.00 ± 0.21, 16.4715 ± 0.0448, and 15.2815 ± 0.0493 across three runs, all consistent with the predicted noiseless value $\dim(\mathrm{Adj}) = 21$ within hardware-noise expectations. Zero-noise extrapolation on the same backend sharpens the cross-group K_9/K_7 ratio test to 3.07% agreement with the structural

prediction $36/21 = 1.714$, bracketing $M(K_9) \in [36, 39]$ at 3σ with the framework's $M = 36$ at the centre. These results support the structural identities underlying both the bracket coefficients in $m_e/m_{\rm Pl}$ and the bit-quantum interpretation of the Schrödinger derivation.

A transverse structural thread is the unification of three *a priori* independent appearances of the integer 168 in NWT: the first Laplacian eigenvalue $\lambda_1(S^3/2I) = 168$ derived in Paper 15 [35]; the order $|\mathrm{PSL}(2,7)| = 168$ of the automorphism group of the Fano plane, shown here to be the symmetry group of the K_7 Wilson amplitude; and the factorisation $168 = 7 \cdot 24 = |\mathrm{Irr}(2T)| \cdot |2T|$ via the McKay correspondence to E_6. These are the same integer, inherited from a single $\mathrm{Spin}(7) \supset 2T$ scaffolding.

1.5 What is derived, identified, and open

We adopt four levels of evidential strength, in order of strength:

Verified experimentally	Measured directly on quantum hardware (IBM Heron R2) with hardware-noise-corrected agreement.
Derived	Forced by the RT framework, Spin(7) representation theory, octonion algebra, or the QEC graph-state framework, up to the Paper 8a input $\alpha = 1/(\sqrt{2}\,\kappa^2)$.
Identified	Matches a structural integer or ratio at CODATA precision, but without a first-principles RT derivation proving the match.
Open	Conjectured or empirically observed, not yet derivable from any of the four frameworks.

With this convention:

- **Derived:** the conservation theorem forcing KE $\rightleftarrows$ topological information (§11.1); the $\sqrt{\alpha}$-per-crossing rule (§3); the $k \leftrightarrow \kappa$ dictionary (§4); the structural requirement of $V \otimes V$ over $\mathrm{Adj}\otimes\mathrm{Adj}$ (§5); the classical prefactor $\dim(S)/\dim(V) = 8/7$ (§6); the graph-state moment identity $\langle H^n_{YY}\rangle_{|K_N\rangle} = \dim(\mathrm{Adj}_{\mathfrak{so}(N)})^n$ for all $n \geq 0$, verified across $N \in \{7, 9, 11\}$ (§11.3–11.5); the octonion sum $\sum |[\cdot]|^2 = 112$ (§9); and the rest-frame Schrödinger evolution $i\hbar\partial_t |K_7\rangle = m_e c^2 |K_7\rangle$ from Bremermann + b2.13 + $\mathrm{PSL}(2,7)$ (§12).

- **Verified experimentally:** the structural identity $\langle H_{YY}\rangle_{|K_7\rangle} = \dim(\mathrm{Adj}_{\mathfrak{so}(7)}) = 21$ on IBM Heron R2 hardware (`ibm_kingston`) in two independent runs: 16.00 ± 0.21 (Run 1, $+75\sigma$ above null) and 16.4715 ± 0.0448 (Run 2, $+368\sigma$ above null), consistent with hardware-noise-corrected prediction (§13). This is the first reproducible experimental verification of the K_7 QEC structure underlying $m_e/m_{\rm Pl}$.

- **Identified:** the α^2 coefficient $\dim(\mathrm{Adj})/\dim(V) = 3$ matches the empirical value to 0.5% (within CODATA precision) via four independent representation- theoretic routes (§8) and is reproduced exactly as the second normalised moment of H_{YY} on $|K_7\rangle$ (§11.4); the per-K_7-vertex reframe of Paper 14's $(1 + \alpha/7)$ as $(1 + \alpha/|V|^2)^{|V|}$ agrees at leading order (§7) and is the first normalised moment on $|K_7\rangle$ (§11.4); the bracket truncates at α^2 because 3-body $\langle Y_a Y_b Y_c\rangle_{|K_7\rangle} = 0$ exactly, ruling out cancellation by higher-body operators (§11.6); the prefactor 8/7 also admits a graph-state Cartan-graded subspace reading (§11.7) which coincides with the Lie reading $\dim(S)/\dim(V)$ at $\mathfrak{so}(7)$ via the rank-3 degeneracy.

- **Open:** a representation-theoretic identification of the α^2-truncation mechanism (Casimir hierarchy / Furry-like theorem [21] / spinor-vector branching) within $\mathrm{Spin}(7)_{k=32}$; a rigorous RT derivation of the α^2 coefficient via an explicit $6j/7j$-symbol calculation on K_7; the extension to α^3; the first-principles origin of the octonion coupling $\lambda = 3/112$ within a specific Wilson amplitude.

This honest decomposition is the spine of the paper: each section's deliverable corresponds to one row in the inventory above.

1.6 Outline

Section 2 sets up the RT Wilson-amplitude framework: Spin(7) Chern–Simons theory at level k, the Heegaard-torus decomposition of $S^3/2I$, and the identification of K_7 as the Wilson graph via Paper 16 b2.13, with explicit provenance from the trefoil $T(2,3)$. Section 3 derives the $\sqrt{\alpha}$-per-edge rule in the RT framework. Section 4 fixes the Chern–Simons level from the Paper 8a vortex aspect ratio. Section 5 proves channel robustness of the per-crossing amplitude. Sections 6 through 9 derive the three coefficients in the bracket of (7) from Spin(7) representation theory and octonion algebra. Section 10 collects the $\mathrm{PSL}(2,7)$ / $\lambda_1 = 168$ unification as a consistency check on the whole framework. Section 11 provides the parallel information-theoretic derivation: the conservation theorem, the K_N stabiliser graph state, the moment identity $\langle H^n_{YY}\rangle = \dim(\mathrm{Adj})^n$, the closed-form lemma $H_{YY} = (M^2 - \dim V)/2$ on the full Hilbert space, the cross-group $\mathfrak{so}(2n+1)$ generalisation, the 3-body truncation result, the three-layer unification of the 8/7 prefactor, and the identification of G as the universal momentum-to-information transduction ratio. Section 12 derives the rest-frame Schrödinger equation $i\hbar\partial_t |K_7\rangle = m_e c^2 |K_7\rangle$ from Bremermann's bound + b2.13 + $\mathrm{PSL}(2,7)$ symmetry, with no additional postulates. Section 13 reports the first experimental verification of the K_7 graph-state structure on IBM's Heron R2 processor, measuring $\langle H_{YY}\rangle = 16.00 \pm 0.21$ consistent with the predicted $\dim(\mathrm{Adj}) = 21$ within hardware noise. Section 14 assembles the closed formula and tests it against CODATA m_e/m_{Pl} and G. Section 15 states the parsimony claim explicitly: NWT's input set reduces to one absolute mass scale plus topological data, with $\hbar$, c, k_B, e as unit conventions. This is the structural minimum allowed by dimensional analysis. Section 16 summarises what is proved, what is identified, and what remains open.

2 Framework: Wilson amplitudes on $S^3/2I$

The amplitude this paper computes is a closed-graph Reshetikhin–Turaev evaluation in three pieces: a 3-manifold, a gauge theory on it, and a graph that carries the matter worldline. This section fixes each piece and the relationships between them.

2.1 The Poincaré sphere and its Heegaard torus

The 3-manifold is the Poincaré homology sphere $M = S^3/2I = \Sigma(2,3,5)$, where $2I$ is the binary icosahedral group of order 120. Equivalently, M is the link of the Brieskorn singularity $x^2+y^3+z^5 = 0$, or the boundary of the E_8 plumbing manifold, or -1 surgery on the right-handed trefoil knot in S^3. Its fundamental group $\pi_1(M) = 2I$ has order 120, with five irreducible $\mathbb{C}$-representations of dimensions $\{1, 2, 2, 3, 3, 4, 6\}$ (binary icosahedral character table).

Among 3-manifolds in the NWT framework, $S^3/2I$ is privileged for two reasons: (a) it is the unique homology 3-sphere with finite fundamental group whose universal cover is S^3, and (b) Paper 15's b2.0 analysis identifies the first non-zero eigenvalue of the Laplace–Beltrami operator on $S^3/2I$ as $\lambda_1 = 168$, the order of $\mathrm{PSL}(2,7)$ (§10).

The manifold admits a Heegaard splitting of genus 1, $M = H_1 \cup_{T^2} H_2$, where each H_i is a solid torus glued along the common boundary T^2 via an element of $\mathrm{SL}(2,\mathbb{Z})$ determined by the $\Sigma(2,3,5)$ surgery data. The Heegaard torus $T^2 \subset M$ is where the K_7 graph lives.

2.2 Spin(7) Chern–Simons theory at level k

The gauge theory is Chern–Simons at level k with gauge group Spin(7). In the Lie-algebra sense this is $\mathfrak{so}(7) = B_3$, rank 3, dual Coxeter number $h^\vee = 5$, with the standard root system in orthogonal basis: simple roots $\alpha_1 = e_1 - e_2$, $\alpha_2 = e_2 - e_3$, $\alpha_3 = e_3$; 9 positive roots; Weyl vector $\rho = (5/2, 3/2, 1/2)$.

The level k is fixed by the $k \leftrightarrow \kappa$ dictionary of §4, which traces back to Paper 8a's Aharonov–Bohm calculation $1/\alpha = 25\pi\sqrt{3} + 1 \approx 137.036$. Numerically $k = 31.73$ at the CODATA value of α, with the integer $k = 32$ closest. Throughout this paper, "$\mathrm{Spin}(7)_k$" means the affine algebra $\widehat{\mathfrak{so}(7)}_k$ at this level, with the quantum group $U_q(\mathfrak{so}(7))$ at $q = e^{i\pi/(k+h^\vee)}$ as the Chern–Simons quantum-group invariant.

2.3 The K_7 graph as the Wilson-amplitude carrier

The matter worldline is a closed Wilson graph $G \subset M$ with prescribed edge labels. Paper 16's b2.13 analysis identifies the carrier graph as the complete graph K_7 on 7 vertices, with 21 edges in canonical bijection with the 21 generators of $\mathfrak{so}(7)$ via the octonion Clifford embedding $2T \hookrightarrow \mathrm{Spin}(7)$ of b2.12 (here $2T \subset 2I$ is the binary tetrahedral subgroup):

$$\{\text{edge } \{i,j\} \in E(K_7)\} \overset{\text{b2.13}}{\longleftrightarrow} \{B_{ij} = L_i L_j : 1 \le i < j \le 7\}. \tag{8}$$

The graph K_7 has a canonical embedding on T^2 via the Heawood map (the unique triangulation of T^2 with K_7 as 1-skeleton, giving 14 triangular faces; Euler characteristic $7 - 21 + 14 = 0$). This places K_7 on the Heegaard torus of M, completing the geometric picture: a Wilson graph whose edges carry the 21 $\mathfrak{so}(7)$ generators, embedded on the Heegaard surface of the Poincaré sphere.

The matter line through K_7 carries the vector representation V of Spin(7) (§5 establishes that the line representation is structurally forced to be V, not the adjoint). The 21 edge labels are insertions of $\mathfrak{so}(7)$ generators along the V-line, in the same way that gluon insertions decorate a quark line in QCD: the matter-line representation and the gauge-boson representation are distinct objects.

2.4 Provenance: from trefoil to K_7

The graph K_7 is not an additional topological input to the NWT framework; it descends from the trefoil $T(2,3)$ alone, via the following chain:

1. The **trefoil** $T(2,3) \subset S^3$ is the only fundamental topological input. It is the simplest non-trivial torus knot and is the carrier of the BPS condensate in Papers 8a [31] and 13 [33].
2. -1 **Dehn surgery** on $T(2,3)$ in S^3 produces the Poincaré sphere $S^3/2I = \Sigma(2,3,5)$. This is the 3-manifold that hosts everything that follows.
3. $S^3/2I$ admits a **genus-1 Heegaard splitting** (§2.1); the splitting torus T^2 is fixed by the Brieskorn structure of $\Sigma(2,3,5)$.
4. By the **Heawood map** ([10], Ringel–Youngs [11]), K_7 admits a unique triangulating embedding on T^2 up to symmetry, with 14 triangular faces.
5. The **b2.13 bijection** [36] matches the 21 edges of K_7 on this embedding to the 21 generators of $\mathfrak{so}(7)$, via the octonion Clifford embedding $2T \hookrightarrow \mathrm{Spin}(7)$ of b2.12.

6. The **graph state** $|K_7\rangle$ on 7 qubits, defined in §11, is the stabiliser state associated to K_7 as a graph in the QEC sense; it carries the Wilson amplitude's information-theoretic content.

Steps 1–5 are geometric; step 6 is the additional QEC layer introduced in this paper and developed in §11.

Terminology disambiguation: K_7 vs the heptafoil

Earlier NWT papers (notably Paper 14 [34], "G from Heptafoil Topology") used the heptafoil knot 7_1 (the $(7,2)$ torus knot, with 7 crossings) as a heuristic carrier for the $\alpha^{21/2}$ scaling. That framing is heuristic, not foundational: under the corrected one-knot reading (Papers 13 and 15), every topological input descends from the trefoil alone, and the integer 7 in $\alpha^{21/2} = (\alpha^{1/2})^{21}$ arises from the 21 edges of K_7 embedded on the Heegaard torus, not from any 7-crossing knot.

Throughout this paper, K_7 denotes the *complete graph* on 7 vertices, with $\binom{7}{2} = 21$ edges, embedded on T^2 via the Heawood map. It is a graph, not a knot, and its 7 vertices and 21 edges are distinct integers from any crossing-number count of any torus knot. The heptafoil framing of Paper 14 is superseded by the present graph-on-Heegaard-torus construction, which descends from the trefoil via steps 1–5 above.

3 The K_7 amplitude and $\sqrt{\alpha}$ per edge

The Wilson amplitude for an electron traversing K_7's Eulerian circuit picks up a factor at each of the 21 edges from the R-matrix of Spin(7) Chern–Simons theory at level k. This section derives that factor from the representation theory of B_3, independently of the $k \leftrightarrow \kappa$ dictionary (which is the subject of §4). The result is a per-edge amplitude $\sin(\pi/(k + h^\vee))$, which under the dictionary becomes $\sqrt{\alpha}$.

3.1 Eulerian circuit on K_7

The complete graph K_7 has $|V(K_7)| = 7$ vertices and $|E(K_7)| = \binom{7}{2} = 21$ edges, every vertex having degree 6. Since all vertices are of even degree and the graph is connected, K_7 admits an Eulerian circuit: a closed walk that traverses each edge exactly once [8]. Hierholzer's algorithm [9] constructs such a circuit explicitly. The Wilson amplitude we compute is the product over these 21 consecutive edge traversals.

The graph K_7 admits a cellular (face-triangulated) embedding on the torus with 14 triangular faces, a construction due to Heawood [10] and completed by Ringel and Youngs [11]. The Euler characteristic is $\chi = |V| - |E| + |F| = 7 - 21 + 14 = 0$, consistent with genus-1 topology. We identify this torus with the Heegaard torus of $S^3/2I$ (§2).

3.2 Wilson lines as $\mathfrak{so}(7)$-valued holonomies

Paper 16 [36] (§b2.13) establishes a bijection between the 21 edges of K_7 and the 21 generators of $\mathfrak{so}(7) = \mathrm{Adj}(\mathrm{Spin}(7))$, lifted via the octonion Clifford embedding $2T \hookrightarrow \mathrm{Spin}(7)$ of Paper 16 b2.12. Each edge $e \in E(K_7)$ therefore carries a Wilson line labelled by the adjoint representation, and the combined amplitude for the Eulerian circuit is a path-ordered product of R-matrix factors at each edge transition.

The Wilson line carrying the electron worldline through the K_7 graph is taken in the vector representation V of Spin(7)—the same representation that hosts the BPS condensate σ-mode of §7 and the Helmholtz eigensolver of Paper 8a [31]. The 21 K_7 edges decorate this V-line with 21 distinct $\mathfrak{so}(7)$ generator insertions B_{ij} under the b2.13 identification [36], but the line's representation is V;

the R-matrix at each edge transition therefore acts on $V \otimes V$. Section 5 shows that the alternative choice $\mathrm{Adj} \otimes \mathrm{Adj}$ does *not* reproduce the per-edge $\sqrt{\alpha}$, fixing V as the canonical line representation.

3.3 The Spin(7) R-matrix in the $V \otimes V$ adjoint channel

In Chern–Simons theory at level k with gauge group G, the R-matrix at a Wilson-line crossing has eigenvalues $\varepsilon_\lambda \, q^{C_\lambda - C_A - C_B}$ on each irreducible sub-representation λ appearing in the fusion $A \otimes B$, where

$$q \;=\; \exp\!\left(\frac{i\pi}{k + h^\vee}\right), \qquad C_\lambda \;=\; \langle \lambda, \lambda + 2\rho \rangle \tag{9}$$

is the quadratic Casimir and $\varepsilon_\lambda = \pm 1$ is $+1$ if $\lambda \subset \mathrm{Sym}^2(V)$ and -1 if $\lambda \subset \Lambda^2(V)$ (Drinfeld normalisation [2]).

For Spin(7) with $h^\vee = 5$, the vector representation V has $\dim V = 7$ and quadratic Casimir $C_V = 6$. The fusion $V \otimes V$ decomposes as

$$V \otimes V \;=\; \mathbf{1} \oplus \mathrm{Adj} \oplus \mathbf{27}, \tag{10}$$

with $\dim(\mathbf{1}) = 1$, $\dim(\mathrm{Adj}) = 21$, and $\dim(\mathbf{27}) = 27$, and Casimirs $C_{\mathbf{1}} = 0$, $C_{\mathrm{Adj}} = 10$, $C_{\mathbf{27}} = 14$. The adjoint sits in $\Lambda^2(V)$, so $\varepsilon_{\mathrm{Adj}} = -1$, and the R-matrix eigenvalue on the adjoint channel is

$$R_{\mathrm{Adj}} \;=\; -\,q^{C_{\mathrm{Adj}} - 2C_V} \;=\; -\,q^{10-12} \;=\; -\,q^{-2}. \tag{11}$$

3.4 The per-edge amplitude $\sin(\pi/(k + h^\vee))$

The amplitude at a single crossing, in the adjoint fusion channel, is the magnitude

$$A_{\mathrm{edge}}(k) \;\equiv\; \frac{|1 + R_{\mathrm{Adj}}|}{2} \;=\; \frac{|1 - q^{-2}|}{2} \;=\; \left|\sin\!\left(\frac{\pi}{k + h^\vee}\right)\right| \;=\; \sin\!\left(\frac{\pi}{k+5}\right). \tag{12}$$

The expansion $|1 - q^{-2}|^2 = 2 - 2\cos(2\pi/(k + h^\vee)) = 4\sin^2(\pi/(k + h^\vee))$ yields the final form.

Equation (12) is the RT per-edge amplitude for a K_7 Wilson line in the adjoint fusion channel of $V \otimes V$. It is a function of the CS level k alone; no appeal to a physical coupling has yet been made.

3.5 Open-path amplitude vs closed-loop probability

Paper 13 [33] (§7.1) establishes a kinematic rule that distinguishes open-path transition amplitudes from closed-loop probability amplitudes on vortex-knot diagrams:

- A *closed loop* at one crossing has probability $|A|^2 \sim \alpha$ per crossing.
- An *open path* at one crossing has amplitude $|A| \sim \sqrt{\alpha}$ per crossing.

The two are related by $|A| = \sqrt{|A|^2}$—i.e., amplitude is the square root of probability, as in ordinary quantum mechanics.

The K_7 Eulerian circuit is traversed by the electron worldline as an *open* path from an in-state to an out-state; it is not a closed loop. Hence the per-edge contribution is the amplitude magnitude (12), not its square. Under the dictionary $\sin(\pi/(k + h^\vee)) = \sqrt{\alpha}$ of §4, (12) becomes $A_{\mathrm{edge}} = \sqrt{\alpha}$, recovering Paper 13's per-crossing amplitude rule.

3.6 Product over 21 edges

The full Eulerian-circuit amplitude is the product of (12) over the 21 edges of K_7:

$$A_{\text{circuit}}(k) \;=\; \left[\sin\!\left(\frac{\pi}{k+h^\vee}\right)\right]^{21}. \tag{13}$$

Under the dictionary of §4, this equals exactly $\alpha^{21/2}$. At CODATA α and the corresponding $k = 31.73$, the numerical value is $\alpha^{21/2} = 3.658 \times 10^{-23}$.

The 21 factors correspond to 21 distinct physical crossings along the Eulerian circuit, each in the adjoint fusion channel. The product (13) is therefore the RT amplitude for an electron worldline passing through the Heegaard-torus K_7-graph structure on $S^3/2I$—the quantity that the remainder of this paper multiplies by the prefactor $\dim(S)/\dim(V)$ and the bracket $[1+\alpha/\dim(V)+\dim(\text{Adj})/\dim(S)\alpha^2]$ to produce m_e/m_{Pl}.

3.7 Remark: representation rigidity

The derivation above is rigid in the choice of representation: the Adj channel of $V \otimes V$ has Drinfeld exponent $C_{\text{Adj}} - 2C_V = -2$, and this exponent is what produces $\sin(\pi/(k+h^\vee))$ via $|1-q^{-2}|/2$. An alternative would take the line in the adjoint, with each edge transition acting on $\text{Adj} \otimes \text{Adj}$. Section 5 shows by direct computation that no irreducible component of $\text{Adj}\otimes\text{Adj}$ carries the exponent $C_\lambda - 2C_{\text{Adj}} = -2$, so the alternative does *not* reproduce the per-edge $\sqrt{\alpha}$. The line representation is therefore fixed to be V, consistent with V hosting the BPS condensate σ-mode of §7.

3.8 Remark: modular naturalness of $\alpha^{21/2}$ for $\mathfrak{so}(7)$

The exponent 21/2 in (13) matches a structural identity in Spin(7) Chern–Simons theory at any level k. Let $\mathcal{D}^2 = \sum_\lambda \dim_q(\lambda)^2$ be the modular total dimension squared, summed over integrable highest weights at level k. At large k (small α), $\mathcal{D}^2$ scales as

$$\mathcal{D}^2 \;\sim\; (k+h^\vee)^{\,\text{rank}(\mathfrak{g})\,+\,2|\Delta_+|} \;=\; (k+h^\vee)^{\,\dim\mathfrak{g}} \tag{14}$$

via $\text{rank}(\mathfrak{g}) + 2|\Delta_+| = \dim\mathfrak{g}$ for any simple Lie algebra.[1] The per-edge amplitude $\sin(\pi/(k+h^\vee)) \sim \pi/(k+h^\vee)$ at large k contributes one factor of $(k+h^\vee)^{-1}$ per K_7 edge, so the full Eulerian-circuit amplitude scales as $(k+h^\vee)^{-|E(K_7)|}$. Combining,

$$\mathcal{D}^2 \cdot \alpha^{|E(K_7)|/2} \;\sim\; (k+h^\vee)^{\,\dim\mathfrak{g}\,-\,|E(K_7)|}. \tag{15}$$

This is level-*independent* (i.e., a finite non-zero limit at $k \to \infty$) precisely when $|E(K_7)| = \dim\mathfrak{g}$. For $\mathfrak{so}(7)$, $|E(K_7)| = 21 = \dim\text{Adj}$, exactly fulfilling this condition by the Paper 16 b2.13 identification of K_7 edges with $\mathfrak{so}(7)$ generators [36]. Numerical verification at levels $k \in \{10, 20, 32, 50, 100\}$ confirms $\mathcal{D}^2 \cdot \alpha^{21/2} \to\approx 2.95 \times 10^{-8}$ at large k.

In other words, the b2.13 bijection $E(K_7) \leftrightarrow \text{Adj}$ does not merely *label* edges by gauge generators; it makes K_7 the unique graph whose RT amplitude on $S^3/2I$ scales modular-naturally—no rank correction or additional structural factor is needed for the amplitude to remain finite as $k \to \infty$ in the appropriate normalisation. This places the choice of K_7 as the carrier graph (Paper 16 b2.13) on a firmer RT-modular footing than the bijection alone provided.

[1] For $B_3 = \mathfrak{so}(7)$: rank $= 3$, $|\Delta_+| = 9$, and $3 + 18 = 21 = \dim\mathfrak{g} = \dim\text{Adj}$.

4 The $k \leftrightarrow \kappa$ dictionary

Section 3 derived the per-edge Wilson amplitude $A_{\text{edge}}(k) = \sin(\pi/(k + h^\vee))$ as a function of the Chern–Simons level k alone. To convert this into a function of the physical fine-structure constant α, we need a dictionary. That dictionary comes from Paper 8a [31], which independently derives α from a Helmholtz eigensolver on the vortex tube. The identification of the two amplitudes at a single crossing fixes k in terms of the NWT vortex aspect ratio $\kappa = R/r$, with the CS level k thereby determined—not a free parameter of the theory.

4.1 Paper 8a's coupling amplitude on the vortex tube

Paper 8a (§3) solves the Helmholtz equation on the surface of the NWT vortex tube, parameterised by torus aspect ratio $\kappa = R/r$ (major-to-minor radius). The $R1$ (electromagnetic) transition operator has matrix element $g_{R1}^2 = 2/\kappa^2$, and the fine-structure coupling is

$$\alpha \;=\; \frac{g_{R1}^2}{2\sqrt{2}} \;=\; \frac{1}{\sqrt{2}\,\kappa^2}. \tag{16}$$

The $1/\sqrt{2}$ factor traces to the amplitude–phase equipartition of the complex condensate field: $m_e^2 = m_{\text{amp}}^2 + m_{\text{phase}}^2 = 2m^{*2}$ with $m^* = m_e/\sqrt{2}$ the effective condensate mass. At $\kappa = \pi^2$ (Paper 8a's preferred value) this gives $\alpha^{-1} = \sqrt{2}\,\pi^4 = 137.76$, within 0.52% of the CODATA value; at $\kappa = \sqrt{1/(\sqrt{2}\,\alpha_{\text{CODATA}})} = 9.8437$ the formula is exact.

The single-crossing amplitude on Paper 8a's vortex tube is the square root of the transition probability,

$$\sqrt{\alpha} \;=\; \frac{1}{2^{1/4}\,\kappa}. \tag{17}$$

4.2 The identification and inversion

Equation (12) gives the RT per-edge amplitude as $\sin(\pi/(k + h^\vee))$; equation (17) gives the same quantity from Paper 8a as $1/(2^{1/4}\,\kappa)$. Identifying the two yields

$$\boxed{\sin\!\left(\frac{\pi}{k + h^\vee}\right) \;=\; \frac{1}{2^{1/4}\,\kappa} \;=\; \sqrt{\alpha}.} \tag{18}$$

Equation (18) is the central dictionary of this paper. It is not a definition: the left-hand side is a function of k derived from Spin(7) representation theory (§3); the right-hand side is a function of κ derived from the Helmholtz eigensolver on the NWT vortex tube [31]. Setting the two equal encodes the physical statement that the RT amplitude at a crossing in the Wilson graph equals the Paper 8a coupling amplitude at a crossing on the vortex tube, i.e., that the two formalisms describe the same physics at each crossing.

Inverting (18):

$$k + h^\vee \;=\; \frac{\pi}{\arcsin\!\big(1/(2^{1/4}\,\kappa)\big)} \;\simeq\; 2^{1/4}\,\pi\,\kappa \;\cdot\; \left[1 + \frac{1}{6\sqrt{2}\,\kappa^2} + O(\kappa^{-4})\right], \tag{19}$$

where the second equality is the large-κ asymptotic. At leading order, $k + h^\vee$ is linear in κ with slope $2^{1/4}\pi$.

4.3 The $2^{1/4}$ factor's physical origin

Four structural factors combine in (18), each with an independent physical interpretation:

Factor	Physical origin
$\sin(\,\cdot\,)$	RT adjoint-channel braiding phase (§3)
$\pi/(k+h^\vee)$	level-shifted Chern–Simons coupling angle
$1/\kappa$	vortex geometry tube-to-torus ratio r/R (Paper 8a [31])
$2^{1/4}$	$\sqrt{m_e/m^*}$, condensate amplitude–phase equipartition [27, 31]

The $2^{1/4}$ is specifically half the logarithmic derivative of the equipartition factor: Paper 8a's $\alpha \propto 1/\sqrt{2}$ folds in the factor $\sqrt{2} = m_e/m^*$, and the dictionary uses the amplitude $\sqrt{\alpha}$, halving the exponent to give $(1/\sqrt{2})^{1/2} = 2^{-1/4}$. Without the equipartition—in a hypothetical NWT condensate with $m^* = m_e$—the dictionary would read $\sin(\pi/(k+h^\vee)) = 1/\kappa$, giving $k + h^\vee \sim \pi\kappa$ at leading order. The factor $2^{1/4}$ is therefore a specific signature of the NWT condensate's amplitude–phase structure, transmitted from Paper 5 [27] and Paper 8a [31] into the RT level determination.

4.4 Numerical evaluation

At the CODATA-exact aspect ratio $\kappa_{\text{CODATA}} = 1/\sqrt{\sqrt{2}\,\alpha_{\text{CODATA}}} = 9.8437$, the dictionary (18) gives

$$k + h^\vee \;=\; \frac{\pi}{\arcsin(1/(2^{1/4}\cdot 9.8437))} \;=\; 36.7314, \qquad k \;=\; 31.7314. \tag{20}$$

At Paper 8a's nominal $\kappa = \pi^2 = 9.8696$ the level is $k = 31.828$. The difference is of order $(\kappa - \pi^2)^2 \sim 10^{-5}$, consistent with the leading-order linearity in κ.

The order of magnitude $k \sim 32$ is physically reasonable for a Chern–Simons gauge theory on a compact 3-manifold: it is not so small that level-truncation effects dominate, nor so large that the classical limit is trivial. That this CS level is determined by the *vortex* aspect ratio, rather than being an independent parameter of the Wilson-amplitude setup, is the content of the dictionary.

4.5 Consequences for subsequent sections

With (18) established, the per-edge amplitude (12) becomes

$$A_{\text{edge}}(\alpha) \;=\; \sqrt{\alpha}, \tag{21}$$

and the Eulerian-circuit product (13) becomes

$$A_{\text{circuit}}(\alpha) \;=\; \alpha^{21/2}. \tag{22}$$

These are the inputs that the remaining sections 5–9 multiply by the prefactor and bracket expansion to produce m_e/m_{Pl}. The dictionary (18) is therefore the single equation that makes all subsequent quantitative claims in the paper expressible in physical rather than CS-level variables.

5 Why $V \otimes V$ and not $\mathrm{Adj} \otimes \mathrm{Adj}$

The $\sqrt{\alpha}$-per-edge rule of §3.3 routes through the adjoint channel of $V \otimes V$, where the Drinfeld R-matrix eigenvalue $-q^{C_{\text{Adj}}-2C_V} = -q^{-2}$ produces $|1+R|/2 = \sin(\pi/(k+h^\vee))$ via the kinematic identity $|1-q^{-2}|^2 = 4\sin^2(\pi/(k+h^\vee))$. This subsection asks whether an alternative line representation—most naturally the adjoint, since the K_7 edges are decorated by $\mathfrak{so}(7)$ generators—would reproduce the same per-edge amplitude through its own fusion. The answer is no: $\mathrm{Adj} \otimes \mathrm{Adj}$ contains no irreducible component with the requisite Drinfeld exponent, fixing V as the canonical line representation.

5.1 The $\mathrm{Adj} \otimes \mathrm{Adj}$ decomposition

For $\mathfrak{so}(7) = B_3$, a from-scratch Freudenthal weight recursion [25] combined with Brauer's tensor-product algorithm [26] yields the 6-component decomposition

$$\mathrm{Adj} \otimes \mathrm{Adj} \;=\; \mathbf{1} \oplus \mathrm{Adj} \oplus \mathbf{27} \oplus \mathbf{35} \oplus \mathbf{168} \oplus (2,1,1)_{189}, \tag{23}$$

with dimension check $1 + 21 + 27 + 35 + 168 + 189 = 441 = 21^2$ and symmetric/antisymmetric split $\mathrm{Sym}^2(\mathrm{Adj}) = \mathbf{1} \oplus \mathbf{27} \oplus \mathbf{35} \oplus \mathbf{168}$ (dim $231 = 21 \cdot 22/2$), $\Lambda^2(\mathrm{Adj}) = \mathrm{Adj} \oplus (2,1,1)_{189}$ (dim $210 = 21 \cdot 20/2$). The 189-dimensional irreducible $(2,1,1)$ is the Spin(7) irrep with highest weight $\omega_1 + 2\omega_3$ in fundamental-weight basis.[2]

5.2 Casimirs and Drinfeld exponents

The quadratic Casimir $C_\lambda = \langle \lambda, \lambda + 2\rho \rangle$ on the six channels of (23) takes the values

$$\begin{array}{c|cccccc} \lambda & \mathbf{1} & \mathrm{Adj} & \mathbf{27} & \mathbf{35} & \mathbf{168} & (2,1,1)_{189} \\ \hline C_\lambda & 0 & 10 & 14 & 12 & 24 & 20 \end{array} \tag{24}$$

With $C_{\mathrm{Adj}} = 10$, the Drinfeld exponent $\xi_\lambda := C_\lambda - 2C_{\mathrm{Adj}}$ on each channel is

$$\xi_\lambda \;\in\; \{-20,\; -10,\; -6,\; -8,\; +4,\; 0\}. \tag{25}$$

The exponent $\xi_\lambda = -2$, which is what would reproduce $|1 - q^{-2}|/2 = \sin(\pi/(k + h^\vee)) = \sqrt{\alpha}$, is absent. Moreover the $(2,1,1)_{189}$ channel sits in $\Lambda^2(\mathrm{Adj})$ and so has Drinfeld sign $\varepsilon = -1$, giving $R_{189} = -q^0 = -1$ exactly: $|1 + R_{189}|/2 = 0$, killing this channel's contribution to leading order.

5.3 Implication: V is structurally required

The Drinfeld exponent $\xi_\lambda = C_\lambda - C_A - C_B = -2$ is what produces the $\sqrt{\alpha}$ amplitude under the dictionary $\sin(\pi/(k + h^\vee)) = \sqrt{\alpha}$ of §4. In $V \otimes V$ this exponent is achieved on the Adj channel, where $C_{\mathrm{Adj}} - 2C_V = 10 - 12 = -2$. In $\mathrm{Adj} \otimes \mathrm{Adj}$, equation (25) shows that no channel achieves this exponent. The smallest-magnitude candidate $|\xi| = 0$ on the $(2,1,1)_{189}$ channel is moreover sign-flipped to give $R = -1$ exactly, which is even further from the linearised regime.

The K_7 Wilson line therefore must carry the vector representation V, not the adjoint. The 21 $\mathfrak{so}(7)$ generator labels of Paper 16 b2.13 [36] on the K_7 edges are decorations of a V-line by Adj-valued insertions, in the same way that gluon insertions decorate a quark line in QCD. The matter-line representation and the gauge-boson representation are distinct. This rigidity is consistent with V being the representation that hosts the BPS condensate σ-mode of §7 and the Helmholtz eigensolver of Paper 8a [31].

6 The classical prefactor $8/7 = \dim(S)/\dim(V)$

The factor 8/7 in Paper 14's formula $m_e/m_{\mathrm{Pl}} = (8/7)\alpha^{21/2}$ is identified as the ratio of two Weyl dimensions in Spin(7), fixed by the spinor–vector branching at the worldline endpoints.

[2] This corrects an earlier listing that placed $\{\mathbf{V}, \mathbf{77}, \mathbf{105}\}$ in $\mathrm{Adj} \otimes \mathrm{Adj}$. Those three irreducibles are real components of Spin(7) but appear in $V \otimes \mathrm{Sym}^2 V = V \oplus \mathbf{77} \oplus \mathbf{105}$ (dim $7 + 77 + 105 = 189 = \dim(2,1,1)$), not $\mathrm{Adj} \otimes \mathrm{Adj}$. Verification by direct Brauer/Klimyk computation in `analysis/nwt_spin7_fusion_compute.py`.

6.1 The two fundamental representations of Spin(7)

Spin(7) has rank 3 and three fundamental representations, indexed by the fundamental weights $\omega_1, \omega_2, \omega_3$:

- The vector representation $V = V_{\omega_1}$, with highest weight $\omega_1 = (1, 0, 0)$ and dimension

$$\dim V \;=\; \prod_{\alpha>0} \frac{\langle \omega_1 + \rho, \alpha \rangle}{\langle \rho, \alpha \rangle} \;=\; 7 \tag{26}$$

  by the Weyl dimension formula on B_3.

- The adjoint representation $\mathrm{Adj} = V_{\omega_2}$, with highest weight $\omega_2 = (1, 1, 0)$ and $\dim \mathrm{Adj} = 21$.

- The spinor representation $S = V_{\omega_3}$, with highest weight $\omega_3 = (1/2, 1/2, 1/2)$ and dimension

$$\dim S \;=\; \prod_{\alpha>0} \frac{\langle \omega_3 + \rho, \alpha \rangle}{\langle \rho, \alpha \rangle} \;=\; 8. \tag{27}$$

The ratio $\dim(S)/\dim(V) = 8/7$ is exact.

6.2 Spinor–vector branching at endpoints

Paper 16's b2.14 derives the appearance of $\dim(S)/\dim(V)$ in the m_e/m_{Pl} amplitude from the boundary structure of the $e \to \sigma$ transition. The electron e carries the spinor representation S of Spin(7) (the Weyl spinor of the BPS fermionic field of Paper 16 §5), while the gauge mode σ carries the vector representation V (the BPS condensate compressibility mode of Paper 8a). The transition amplitude $\langle \sigma | A | e \rangle$ therefore involves the projector

$$\Pi_{S \to V} \;:\; S \otimes V^* \;\to\; (\text{intertwiner space}), \tag{28}$$

whose normalisation contributes a classical factor $(\dim S \cdot \dim V)^{-1/2}$ at each endpoint. Paper 16 b2.14 shows that the precise combinatorial factor that emerges from the two endpoints (one for e, one for σ) is

$$\frac{\dim(S)}{\dim(V)} \;=\; \frac{8}{7}, \tag{29}$$

multiplying the bulk amplitude $\alpha^{21/2}$ from the 21 edges.

6.3 Why not the quantum-dimension ratio

A naive alternative would be to use the quantum dimensions $\dim_q(S)/\dim_q(V)$ at level k, which include α corrections. Phase 6.4 evaluates this ratio at level $k = 32$:

$$\frac{\dim_q(S)}{\dim_q(V)} \;=\; \frac{8}{7}\left[1 + \frac{5\alpha}{8} + O(\alpha^2)\right], \tag{30}$$

where the $5\alpha/8$ correction follows from Freudenthal's strange formula $\Delta(\lambda) = h^\vee \cdot C(\lambda)$ applied to S and V: $h^\vee(C(V) - C(S))/6 = 5 \cdot (6 - 21/4)/6 = 5/8$.

The NLO coefficient $5\alpha/8$ does *not* match Paper 14's NLO coefficient $\alpha/7$. Hence the $\alpha/7$ NLO does not arise from quantum-dimension renormalisation of the prefactor, and the 8/7 factor is taken at its classical value. The $\alpha/7$ NLO has a different structural source—a per-vertex correction on the K_7 graph—developed in §7 below, and re-derived as the first normalised moment of H_{YY} on $|K_7\rangle$ in §11.4.

6.4 Cross-check: the K_7 Cartan-graded subspace

The classical prefactor 8/7 also has a graph-state interpretation that complements the spinor–vector boundary projection above and will be developed in §11.7. Briefly: fixing $N - \mathrm{rank}(\mathfrak{so}(N)) = 7-3 = 4$ stabilisers of $|K_7\rangle$ at $+1$ produces an exact $2^{\mathrm{rank}(\mathfrak{so}(7))} = 8$-dimensional "Cartan-graded" subspace, with $|K_7\rangle$ inside and a 7-dimensional non-singlet complement. The trace ratio $\mathrm{Tr}(\Pi_S)/\mathrm{Tr}(\Pi_S - |K_7\rangle\langle K_7|) = 8/7$ matches the spinor-vector dimension ratio. At rank 3, this is the same number as the spinor–vector boundary projection $\dim(S)/\dim(V) = 8/7$.

The graph-state and Lie readings happen to coincide at $\mathfrak{so}(7)$ because of the rank-3 degeneracy $\dim(V_{\mathfrak{so}(7)}) = 7 = 2^{\mathrm{rank}(\mathfrak{so}(7))} - 1 = \dim(S_{\mathfrak{so}(7)}) - 1$; the readings would give different numbers for any $\mathfrak{so}(2n+1)$ with $n \geq 4$, and the spinor–vector boundary projection is the physically motivated identification.

7 The per-vertex $(1 + \alpha/|V|^2)^{|V|}$ form

Paper 14 introduces an NLO factor $(1 + \alpha/7)$ multiplying $\alpha^{21/2}$ in m_e/m_{Pl}. Section 6 ruled out the quantum-dimension ratio as its source. This section identifies an alternative structural source: a per-vertex self-energy on the K_7 Wilson graph, with $|V(K_7)| = 7$ vertices each contributing an $\alpha/49$ correction.

7.1 Binomial reframing of Paper 14's NLO

Paper 14's $(1 + \alpha/7)$ admits a binomial reframing:

$$\left(1 + \frac{\alpha}{49}\right)^7 \;=\; 1 \;+\; \binom{7}{1}\frac{\alpha}{49} \;+\; \binom{7}{2}\frac{\alpha^2}{49^2} \;+\; \cdots \;=\; 1 \;+\; \frac{\alpha}{7} \;+\; \frac{21\,\alpha^2}{2401} \;+\; O(\alpha^3). \tag{31}$$

At leading order this matches Paper 14's NLO exactly. The α^2 contribution from the binomial is $21\alpha^2/2401 \approx 8.74 \times 10^{-3}\,\alpha^2$, which is much smaller than the $3\alpha^2$ NNLO coefficient of §8 (and is absorbed into the $O(\alpha^3)$ truncation error at CODATA precision). The binomial form is therefore a valid *per-vertex* interpretation of the NLO factor.

7.2 Identification with K_7 vertices

For (31) to be more than a numerical coincidence, the integers 7 and 49 must have direct structural interpretations. The K_7 Wilson graph supplies both:

$$|V(K_7)| \;=\; 7 \;=\; \dim V_{\mathrm{Spin}(7)}, \qquad |V(K_7)|^2 \;=\; 49 \;=\; \dim(V)^2. \tag{32}$$

The first equality is the b2.13 identification of the 7 K_7 vertices with the 7 basis vectors of the vector representation V of Spin(7) (specifically, with the seven imaginary octonions, via the octonion Clifford embedding $2T \hookrightarrow \mathrm{Spin}(7)$). The second is the squared counterpart that appears in the per-vertex denominator.

With the identifications (32), the binomial (31) reads

$$\left(1 + \frac{\alpha}{\dim(V)^2}\right)^{\dim(V)} \;=\; 1 \;+\; \frac{\alpha}{\dim(V)} \;+\; O(\alpha^2) \tag{33}$$

—a manifestly representation-theoretic form, with no graph- specific data appearing on the right-hand side beyond the dimension of the vector representation.

7.3 Per-vertex self-energy interpretation

Equation (33) suggests a physical interpretation: each of the 7 K_7 vertices contributes an independent NLO correction $\alpha/\dim(V)^2$, and the total NLO is the binomial product over vertices. The factor $1/\dim(V)^2$ has the right dimensional content for a "self-energy" on a single vertex: $\dim(V)^2$ is the number of $V \otimes V$ matrix elements at a vertex (without any selection rule), and α is the natural dimensionless coupling at level k.

This is consistent with the Paper 16 b2.13 picture of K_7 as a spin-network on the Heegaard torus: at each vertex, six V-edges meet, and the local intertwiner amplitude has an α-correction controlled by the Casimir balance of the $V \otimes V$ fusion at the vertex. The per-vertex prediction $\alpha/\dim(V)^2 = \alpha/49$ is what the binomial expansion gives; whether this is an exact RT result or an effective identification at NLO is the topic of §16.4.

7.4 Caveat: this is an identification, not a derivation

Phase 7 of the NWT program attempted three independent RT derivations of $(1 + \alpha/49)^7$ from $\mathrm{Spin}(7)_{k=32}$ data (§16.4): a Schur–Racah reduction of the K_7 spin network, a Heegaard modular-S ansätz, and a Lawrence–Rozansky surgery sum [6] on the trefoil. None reproduces the per-vertex coefficient $\alpha/49$ from a single direct manipulation.

We therefore present (33) as a *structural identification*: the NLO factor matches the Spin(7) representation ratio $|V(K_7)|^{-2}$ summed binomially over the 7 K_7 vertices, which itself matches by the b2.13 bijection. The first-principles RT derivation of the per-vertex $\alpha/49$ remains the central target of the bracket-derivation program outlined in §16.4.

7.5 Cross-check: graph-state first moment

The information-theoretic derivation in §11 provides an independent route to the same NLO coefficient. The identity $\langle H_{YY}\rangle_{|K_7\rangle} = \dim(\mathrm{Adj}_{\mathfrak{so}(7)}) = 21$ combined with the natural normalisation $1/[\dim(V)\,\dim(\mathrm{Adj})]$ gives

$$\frac{\alpha\,\langle H_{YY}\rangle_{|K_7\rangle}}{\dim(V)\cdot\dim(\mathrm{Adj})} = \frac{\alpha\cdot 21}{7\cdot 21} = \frac{\alpha}{7} = \frac{\alpha}{\dim V}, \tag{34}$$

matching (33) exactly. The graph-state moment identity is structurally distinct from the per-vertex binomial (33)—the NLO arises as a single first-moment expectation value on a 128-dimensional Hilbert space, not as a binomial product over vertices—but the two routes converge on the same coefficient, providing cross-confirmation that $\alpha/\dim(V) = 1/7$ is the correct structural identification.

8 The α^2 coefficient $\dim(\mathrm{Adj})/\dim(V)$

This section derives the α^2 coefficient in the bracket of (7). The starting point is an *empirical* residual from Paper 14's formula; the ending point is a structural identification with a Spin(7) representation ratio that appears, independently, via four distinct routes. The Paper 16 b2.13 identification of K_7 with Spin(7) makes these routes commensurate rather than independent observations of the same integer.

8.1 The α^2 residual from Paper 14

Paper 14's formula $m_e/m_{\rm Pl} = (8/7)(1+\alpha/7)\,\alpha^{21/2}$ agrees with CODATA $m_e/m_{\rm Pl}$ to -0.014%. The residual corresponds to a specific α^2 coefficient c_2 in an extended prefactor of the form

$$P(\alpha) \;\equiv\; \frac{m_e/m_{\rm Pl}}{\alpha^{21/2}} \;=\; c_0 + c_1\alpha + c_2\alpha^2 + O(\alpha^3), \tag{35}$$

where $c_0 = 8/7$ and $c_1 = (8/7)(1/\dim V) = 8/49$ are the Paper 14 values from §6 and §7. Using CODATA values $\alpha = 7.2973525693\times10^{-3}$ and $m_e/m_{\rm Pl} = 4.18544\times10^{-23}$, the required α^2 coefficient is

$$c_2^{\rm empirical} \;=\; \frac{m_e/m_{\rm Pl}}{\alpha^{21/2}} - \frac{8}{7} - \frac{8}{49}\,\alpha \Big/ \alpha^2 \;=\; 3.01 \pm 0.2. \tag{36}$$

The uncertainty is set by CODATA $m_e/m_{\rm Pl}$, which is itself limited by the 22 ppm uncertainty on G (§14.4).

8.2 Structural identification

Within the representation theory of Spin(7) $= B_3$, the integer nearest (36) that has a structural interpretation is

$$c_2 \;=\; \frac{\dim \mathrm{Adj}}{\dim V} \;=\; \frac{21}{7} \;=\; 3, \tag{37}$$

or equivalently, in bracket form,

$$\delta \;\equiv\; \frac{c_2}{c_0} \;=\; \frac{\dim \mathrm{Adj}}{\dim S} \;=\; \frac{21}{8} \;=\; 2.625. \tag{38}$$

Empirical $c_2 = 3.01$ matches (37) to 0.5%, well inside the $\pm 7\%$ window allowed by the CODATA precision on $m_e/m_{\rm Pl}$ (§14.4).

8.3 Four independent routes to the integer 3

The integer 3 in (37) is not a coincidence. It appears via four routes, each anchored to a different piece of the Spin(7)/$2T/E_6$ scaffolding, each giving the same value.

Route 1: Spin(7) irreducible representation ratio. By direct evaluation, $\dim \mathrm{Adj} = 21 = \binom{7}{2}$ (antisymmetric 2-tensor of V) and $\dim V = 7$, so $\dim \mathrm{Adj}/\dim V = 3$. This is the most direct identification and what (37) states.

Route 2: Rank of $\mathfrak{so}(7)$. The rank of B_3 is the dimension of its Cartan subalgebra, which counts independent diagonal generators. For $\mathfrak{so}(2n+1)$ the rank is n, so $\mathrm{rank}(\mathfrak{so}(7)) = 3$. This is a purely Lie-theoretic quantity and, for Spin(7) specifically, it equals $\dim \mathrm{Adj}/\dim V$. The equivalence is the general formula

$$\mathrm{rank}(\mathfrak{so}(2n+1)) \;=\; n \;=\; \frac{(2n+1)-1}{2} \;=\; \frac{\dim \mathrm{Adj}(\mathfrak{so}(2n+1))}{\dim V(\mathfrak{so}(2n+1))}. \tag{39}$$

(Here $\dim \mathrm{Adj} = n(2n+1)$ and $\dim V = 2n+1$.)

Route 3: The unique 3-dimensional irrep of $2T$. The binary tetrahedral group $2T$ has order 24 and exactly seven irreducible complex representations with dimensions

$$\{1, 1, 1, 2, 2, 2, 3\}. \tag{40}$$

The 3-dimensional irrep is the unique non-2-dimensional non-trivial representation. Its dimension 3 equals the α^2 coefficient. The connection to our construction is through Paper 16 b2.12, which shows that $2T \hookrightarrow \mathrm{Spin}(7)$ via the octonionic Clifford structure on $\mathrm{Cl}(0, 7)$, so the $2T$ irrep theory is not tangential but is the discrete quotient that generates $S^3/2I$.

Route 4: Central Coxeter label of E_6. Under the McKay correspondence, the binary tetrahedral group $2T$ is dual to the exceptional Lie algebra E_6, with the irreps of $2T$ indexed by nodes of the extended E_6 Dynkin diagram. The Coxeter labels (equivalently, dimensions of the McKay irreps) on that diagram are

$$1, 2, 3, 2, 1, 2, 1, \qquad \text{sum } = h_{E_6} = 12, \tag{41}$$

with the central triply-connected node carrying label 3. The α^2 coefficient thus equals the unique branching label of the E_6 Dynkin diagram.

These four routes are not independent pieces of evidence in the Bayesian sense—they follow from a single $\mathrm{Spin}(7) \supset 2T$ scaffolding—but they show that $3 = c_2$ is fixed by the representation theory at multiple levels of that scaffolding. If any one route gave a different answer, the match in (37) would be accidental; the fact that all four agree is what promotes (37) from a numerical observation to a structural identification.

8.4 A negative result: the quantum-dim ratio is not the NLO source

It is tempting to identify the Paper 14 NLO $(1 + \alpha/7)$ with the quantum-dim ratio $\dim_q(S)/\dim_q(V)$ at level k, since both equal 8/7 in the classical ($k \to \infty$) limit. The Kac–Weyl formula gives, for Spin(7) at level k with $q = \exp(i\pi/(k + h^\vee))$,

$$\frac{\dim_q(S)}{\dim_q(V)} = \frac{[1]_q^2\,[3]_q}{[7/2]_q\,[3/2]_q\,[1/2]_q}, \qquad [n]_q \equiv \frac{\sin(n\pi/(k + h^\vee))}{\sin(\pi/(k + h^\vee))}. \tag{42}$$

Expanding in the small parameter $\alpha = \sin^2(\pi/(k + h^\vee))$ (Paper 8a dictionary, §4),

$$\frac{\dim_q(S)}{\dim_q(V)} = \frac{8}{7}\left(1 + \frac{5\alpha}{8} + O(\alpha^2)\right). \tag{43}$$

The NLO coefficient is **5/8**, not **1/7**. Numerically, using the quantum-dim ratio as the prefactor overshoots CODATA m_e/m_{Pl} by +0.45%—far outside Paper 14's 0.014% envelope. The Paper 14 $(1 + \alpha/7)$ is therefore *not* of quantum-dim origin.

The α^2 coefficient $c_2 = \dim(\mathrm{Adj})/\dim(V)$ from (37) is a structurally and numerically distinct quantity from the α and α^2 terms in (42). Its α-level coefficient from the quantum-dim route ($5/8 = 0.625$) differs from the Paper 14 empirical value ($1/7 \approx 0.143$) by a factor of 4.4; its α^2-level coefficient (computed numerically from (42) at $k = 31.73$) is approximately 0.93, not the empirical $c_2 = 3.01$. Both components of the quantum-dim ratio are therefore ruled out as the primary NLO/NNLO source.

8.5 Generalisation across Spin(N)

The structural identification (37) extends to $\mathfrak{so}(N)$ for general odd N:

$$c_2(\mathfrak{so}(N)) \;=\; \frac{\dim \mathrm{Adj}(\mathfrak{so}(N))}{\dim V(\mathfrak{so}(N))} \;=\; \frac{N-1}{2} \;=\; \mathrm{rank}(\mathfrak{so}(N)). \tag{44}$$

For the NWT-relevant case $N = 7$, this gives $c_2 = 3$ and recovers (37). The generalisation is a testable prediction: a hypothetical NWT construction with $\mathfrak{so}(9)$ gauge group would have $c_2 = 4$; with $\mathfrak{so}(11)$, $c_2 = 5$; and so on. We do not construct such extensions here, but note that the formula (102) would change its numerical output accordingly, providing a sharp falsification test should an independent motivation for a different N emerge.

8.6 Summary

(37) identifies the α^2 prefactor coefficient with the Spin(7) rank via four independent representation-theoretic routes, with 0.5% numerical agreement against CODATA (within the current 7% precision limit). This identification is epistemically *identified* rather than *derived* in the sense of §1.5: the numerical match holds, and four routes converge, but a direct first-principles RT $6j/7j$-symbol calculation establishing $c_2 = \dim(\mathrm{Adj})/\dim(V)$ from the K_7 Wilson amplitude alone remains open. §9 closes part of that gap by showing that $c_2 = 3$ follows from an octonion-associator sum rule if the per-triangle amplitude is proportional to $|[e_a, e_b, e_c]|^2$—providing an *independent* route to the same integer via the octonion composition-algebra structure underlying Spin(7).

9 Octonion derivation of $\sum |[\,\cdot\,]|^2 = 112$

Section 8 identified the α^2 coefficient with a Spin(7) representation ratio on *empirical* and *structural* grounds. The octonion structure of the 7-dimensional vector representation of Spin(7) offers an *algebraic* route to the same coefficient. This section carries out the derivation in four steps: the octonion algebra on the 7 imaginary basis elements (§9.1); the Cartan 3-form and its values on unordered basis triples (§9.2); the associator $[e_a, e_b, e_c] = (e_a e_b)e_c - e_a(e_b e_c)$ and the composition-algebra identity for its squared norm (§9.3); and the summation over basis triples giving $\sum |[e_a, e_b, e_c]|^2 = 112$ (§9.4). The section closes in §9.5 by showing that this octonion sum, combined with the per-triangle amplitude hypothesis of Paper 16 b2.15, gives an *independent* route to $c_2 = \dim(\mathrm{Adj})/\dim(V) = 3$.

9.1 The seven imaginary octonions and the Fano plane

Write $\mathbb{O} = \mathbb{R} \oplus \mathrm{Im}(\mathbb{O})$ for the real octonion algebra, with $\mathrm{Im}(\mathbb{O})$ spanned by seven orthonormal imaginary units $e_1, \ldots, e_7$ satisfying

$$e_i^2 = -1 \quad (i = 1, \ldots, 7), \qquad e_i e_j = -e_j e_i \quad (i \neq j). \tag{45}$$

The non-trivial products $e_i e_j$ for $i \neq j$ are encoded by seven *Fano lines*, each a cyclic 3-tuple (a, b, c) on which $e_a e_b = e_c$, $e_b e_c = e_a$, $e_c e_a = e_b$. In the standard (so-called "natural cyclic") convention the seven lines are

$$\{(1,2,4),\ (2,3,5),\ (3,4,6),\ (4,5,7),\ (5,6,1),\ (6,7,2),\ (7,1,3)\}. \tag{46}$$

Each pair $\{a, b\} \subset \{1, \ldots, 7\}$ is contained in exactly one Fano line, so (45) and (46) completely determine the multiplication table on $\mathrm{Im}(\mathbb{O})$.

As an unordered combinatorial object, the set $\{1, \ldots, 7\}$ with the Fano lines as 3-subsets forms the Fano plane $\mathrm{PG}(2, \mathbb{F}_2)$, the projective plane over $\mathbb{F}_2$. The 7 points and 7 lines come from identifying

e_i with the nonzero vectors of $\mathbb{F}_2^3$ and lines with triples that sum to zero; the oriented multiplication convention picks one of the two orientations of each line.

9.2 The Cartan 3-form φ

The octonion structure constant

$$f^{abc} \;=\; \begin{cases} +1 & \text{if } (a,b,c) \text{ is a cyclic Fano line,} \\ -1 & \text{if } (a,b,c) \text{ is an anti-cyclic Fano line,} \\ 0 & \text{otherwise} \end{cases} \tag{47}$$

is totally antisymmetric in (a,b,c) and invariant under the exceptional group $G_2 = \mathrm{Aut}(\mathbb{O})$. We write

$$\varphi(a,b,c) \;=\; f^{abc}\, a_a\, b_b\, c_c \qquad \text{(sum over repeated indices)} \tag{48}$$

for its action on three imaginary octonions. On orthonormal basis vectors, $\varphi(e_a, e_b, e_c) = \pm 1$ exactly when $\{a,b,c\}$ is a Fano line (and zero otherwise), with the sign fixed by orientation relative to (46). In particular

$$|\varphi(e_a, e_b, e_c)|^2 \;=\; \begin{cases} 1 & \{a,b,c\} \text{ is a Fano line,} \\ 0 & \text{otherwise.} \end{cases} \tag{49}$$

Among the $\binom{7}{3} = 35$ unordered triples of distinct indices, exactly 7 are Fano lines and $35 - 7 = 28$ are "non-Fano" (three distinct indices whose pairs lie on three *different* Fano lines).

9.3 The associator and its squared norm on basis triples

Because $\mathbb{O}$ is non-associative, the trilinear associator

$$[a,b,c] \;\equiv\; (ab)c - a(bc) \tag{50}$$

is nonzero in general. On $\mathrm{Im}(\mathbb{O})$ the associator is totally antisymmetric (so it vanishes whenever two arguments coincide) and G_2-invariant. Baez [12, §4.1] observes that the Cartan 3-form φ, the G_2-invariant cross product $a \times b$, and the associator $[a,b,c]$ are "almost the same thing": the associator is, up to a constant, the dualisation of the Hodge dual ψ of φ. This structural identity yields a numerical one on orthonormal basis triples.

For $a = e_a, b = e_b, c = e_c$ imaginary octonion basis vectors with $a \neq b \neq c$, direct computation of $(e_a e_b)e_c - e_a(e_b e_c)$ using the multiplication table generated by (46) gives

$$|[e_a, e_b, e_c]|^2 \;=\; \begin{cases} 0 & \{a,b,c\} \text{ is a Fano line,} \\ 4 & \{a,b,c\} \text{ is non-Fano.} \end{cases} \tag{51}$$

This can be written uniformly as

$$|[e_a, e_b, e_c]|^2 \;=\; 4\big(1 - |\varphi(e_a, e_b, e_c)|^2\big), \tag{52}$$

using (49). Equation (52) is the restriction to orthonormal basis triples of a more general composition-algebra identity $|[a,b,c]|^2 = 4(\det\langle a_i, a_j\rangle - |\varphi(a,b,c)|^2)$ that holds for arbitrary imaginary octonions; the Gram determinant reduces to 1 on orthonormal triples, recovering (52). We verify the orthonormal form directly on all 35 basis triples rather than derive the general form.

The associator vanishes on Fano triples because such triples generate a quaternion subalgebra of $\mathbb{O}$, and quaternions are associative. On non-Fano triples the associator is a nonzero octonion of squared norm 4; direct computation shows $[e_a, e_b, e_c] = \pm 2\, e_d$ for some $d \notin \{a,b,c\}$. The sign is fixed by the chosen orientation in (46); the squared norm does not depend on the orientation.

9.4 Summation over basis triples

Summing (51) over the $\binom{7}{3} = 35$ unordered triples:

$$\sum_{\{a,b,c\}} |[e_a, e_b, e_c]|^2 \;=\; 4\sum_{\{a,b,c\}} 1 - 4\sum_{\{a,b,c\}} |\varphi|^2 \;=\; 4\cdot\binom{7}{3} - 4\cdot|\{\text{Fano lines}\}| \;=\; 4\cdot 35 - 4\cdot 7 \;=\; \boxed{112}. \quad (53)$$

Equivalently, $\sum |[e_a, e_b, e_c]|^2 = 4 \cdot 28 = 4 \cdot |\{\text{non-Fano triples}\}|$.

Both factors are structural: the 4 is fixed by the composition identity (52) through the orthonormality of the basis, and the 28 is $\binom{|V|}{3} - |\text{Fano}|$ for $|V| = 7$.

Equation (53) has been verified by direct computation on the multiplication table generated by (46), covering all 35 triples: the 7 Fano triples give associator 0 exactly, and the 28 non-Fano triples each have $|[\,\cdot\,]|^2 = 4$ exactly.

The arithmetic identity $112 = 8\cdot 14 = \dim(S)\cdot|F(K_7)|$, with $|F(K_7)| = 14$ the number of triangular faces of the Heawood embedding of K_7 on the torus [11], is a coincidence of elementary arithmetic rather than an additional structural statement: $4 \cdot 28 = 8 \cdot 14$ because $\binom{7}{3} - 7 = 2 \cdot |F|$ for the Heawood embedding, not because of an independent count.

9.5 From the octonion sum to $c_2 = 3$

Paper 16 (§b2.15) hypothesises that the α^2 contribution to the prefactor $P(\alpha)$ of (35) decomposes as a sum over K_7 triangles weighted by the squared associator of their vertices:

$$c_2 \;=\; \lambda \sum_{\{a,b,c\}\subset V(K_7)} |[e_a, e_b, e_c]|^2 \;=\; \lambda \cdot 112, \quad (54)$$

where λ is a structural coupling. Combining (54) with the Spin(7) identification $c_2 = \dim(\text{Adj})/\dim(V) = 3$ from §8 fixes

$$\lambda \;=\; \frac{c_2}{\sum |[\,\cdot\,]|^2} \;=\; \frac{\dim(\text{Adj})/\dim(V)}{4\big(\binom{|V|}{3} - |\text{Fano}|\big)} \;=\; \frac{3}{112} \;=\; \frac{\dim(\text{Adj})}{4 \cdot \dim(V) \cdot |\{\text{non-Fano triples}\}|}. \quad (55)$$

Zero free parameters: λ is completely determined by the gauge-algebra irrep dimensions and the Fano-plane count.

The per-triangle amplitude hypothesis (54) is retained from Paper 16 as a structural assumption and is not proved from the RT amplitude directly (a rigorous $6j/7j$-symbol derivation on K_7 remains an open problem—see §16). What (53)–(55) establish is that this hypothesis is *consistent* with the Spin(7) structural identification of §8, and that both point to the same integer $c_2 = 3$ via fully independent routes (Spin(7) representation theory for §8; octonion composition algebra plus Fano counting for this section).

9.6 Scope and limitations

The derivation of (53) depends on two structural inputs: the basis-triple identity (52), which is specific to $\mathbb{O}$ through the Cartan 3-form's support on Fano lines and is a consequence of the octonion composition-algebra structure described in [12, §4.1]; and the Fano-plane count $|\text{Fano}| = 7 = |V|$, which is specific to $\text{PG}(2, \mathbb{F}_2)$. Together these make the octonion route specific to Spin(7) and not transferable to Spin(N) for $N \neq 7$ without an independent motivation for a Fano-like structure on the vector rep.

This specificity is consistent with the generalised identification $c_2 = \text{rank}(\mathfrak{so}(N))$ from §8.5: the Spin(7) octonion route is one particular derivation of the rank identity, but the identity itself is more general. For other $\mathfrak{so}(N)$ a different algebraic structure would underlie any analogous result.

10 $\mathrm{PSL}(2,7)$ equivariance and the 168 unification

The integer 168 appears at four independent points of the construction. This brief section assembles them and traces all four to the same Spin(7)$\rightarrow 2T \rightarrow E_6$ scaffolding.

10.1 Four occurrences of 168

The four occurrences are:

1. $|\mathrm{PSL}(2,7)| = 168$, the order of the simple group of $\mathbb{F}_7$-projective 2×2 matrices. By a classical isomorphism, $\mathrm{PSL}(2,7) \cong \mathrm{GL}(3,\mathbb{F}_2)$, the automorphism group of the Fano plane (the projective plane over $\mathbb{F}_2$, consisting of 7 points and 7 lines with 3 points per line). $\mathrm{PSL}(2,7)$ acts edge-transitively on K_7 (every edge in the same $\mathrm{PSL}(2,7)$-orbit) and partitions the $\binom{7}{3} = 35$ unordered triples of K_7 vertices into two orbits: 7 "Fano triangles" (the lines of the Fano plane) and 28 "non-Fano triangles".

2. $\lambda_1(S^3/2I) = 168$, the first non-zero eigenvalue of the Laplace–Beltrami operator on the Poincaré sphere (Paper 15 b2.0 [35]). This is established via the Fadell–Pisani character formula on $S^3/2I$ and corresponds to a specific $2I$-equivariant scalar mode.

3. $|\mathrm{Irr}(2T)| \cdot |2T| = 7 \cdot 24 = 168$, where $\mathrm{Irr}(2T)$ has 7 irreducible characters and $|2T| = 24$ is the order of the binary tetrahedral group. Via the McKay–E_6 correspondence, the 7 irreducible $2T$-reps are in canonical bijection with the 7 nodes of the affine E_6 Dynkin diagram, and the 24 elements of $2T$ are in canonical bijection with the trefoil's $2I$-orbit on the Heegaard torus (stabiliser $\mathbb{Z}_5$ in $2I$).

4. In a $\mathrm{Spin}(7)_k$ Wilson amplitude on K_7, the $\mathrm{PSL}(2,7)$ subgroup acts as the symmetry of the carrier graph (Phase 6.7), constraining the $6j$-coefficients of the spin-network evaluation to be $\mathrm{PSL}(2,7)$-equivariant. The size of this equivariance group is $|\mathrm{PSL}(2,7)| = 168$.

10.2 All four descend from one scaffolding

The Paper 16 b2.12–b2.13 chain identifies a single chain of inclusions:

$$2T \hookrightarrow 2I \hookrightarrow \mathrm{SU}(2), \qquad 2T \hookrightarrow \mathrm{Spin}(7) \ \text{(via Cl(0,7) octonion Clifford)}, \tag{56}$$

with the E_6 McKay correspondence linking $2T$ and the affine E_6 Dynkin diagram. The integer 168 arises in each of the four occurrences above as a consequence of this scaffolding:

- $|\mathrm{PSL}(2,7)|$: $\mathrm{PSL}(2,7) = \mathrm{GL}(3,\mathbb{F}_2)$ is the symmetry group of the Fano plane, which arises in Spin(7)'s octonion algebra structure (the seven imaginary octonions plus their multiplication table form the Fano plane; §9).

- $\lambda_1(S^3/2I) = 168$: the eigenvalue is fixed by the $2I$ representation theory on $S^3 = \mathrm{SU}(2)$, which inherits E_6-related multiplicities through the $2T \subset 2I$ inclusion.

- $|\mathrm{Irr}(2T)| \cdot |2T| = 7 \cdot 24$: the 7 from E_6 Dynkin nodes, the 24 from the $2T$-orbit of the trefoil on the Heegaard torus.

- $\mathrm{Spin}(7)_k$ K_7 equivariance: $\mathrm{PSL}(2,7)$ is the automorphism group of the Fano plane, hence of the seven imaginary octonions, hence of the seven K_7 vertices via b2.13.

The four occurrences of 168 are therefore not coincidental but manifestations of a single piece of structure (Spin(7)$\to 2T \to E_6$). This is the strongest cross-domain consistency check of the Paper 17 framework: an integer that appears as a Laplace eigenvalue on a 3-manifold, as a discrete group order, as a $2T$-character-table count, and as a Wilson-graph symmetry group order, all coincides at 168 because each is controlled by the same scaffolding.

11 The QEC graph-state derivation of the bracket

The preceding sections derived each factor in m_e/m_{Pl} from Spin(7) representation theory, the Reshetikhin–Turaev R-matrix, and octonion algebra. This section provides an *information-theoretic* derivation of the bracket $[1 + \alpha/\dim(V) + \dim(\mathrm{Adj})/\dim(V)\,\alpha^2]$ by recognising that the integers in those bracket coefficients are exactly the moments of a specific operator on the K_7 stabiliser graph state.

The result has three pieces: (i) a structural identity $\langle H^n_{YY}\rangle_{|K_N\rangle} = \dim(\mathrm{Adj}_{\mathfrak{so}(N)})^n$ valid for all $n \geq 0$ and verified across the $\mathfrak{so}(2n+1)$ family at $N \in \{7, 9, 11\}$; (ii) a 3-body operator probe showing the bracket truncates at α^2 for representation-theoretic, not cancellation, reasons; and (iii) the identification of G as the universal transduction ratio between momentum-changing interaction events and topological-information creation, which the K_7 graph-state derivation makes quantitative.

11.1 The information conservation theorem

The information-theoretic framing rests on a conservation theorem that is independent of any specific dynamical mechanism. Energy conservation in a closed system (agent + particle + medium) requires that any kinetic-energy change of an accelerating particle transfer to its surroundings. In a superfluid medium with $\eta = 0$, heat dissipation is forbidden. The remaining sinks for energy are the medium's configurational degrees of freedom, which carry information in the Bekenstein–Shannon sense [19, 20]. The conservation theorem is then

$$\boxed{\Delta(\mathrm{KE}) \;\rightleftarrows\; \Delta(\text{topological information})} \tag{57}$$

forced by energy conservation and zero superfluid viscosity, prior to any specific quantitative implementation (Verlinde, Whitehead- occasions, or otherwise).

For *stable topological solitons* (NWT particles), the configurational-DOF sharpens further: continuous medium response (density profile, phase gradient, bulk flow) advects with the particle as it propagates, so only the *discrete* topological invariants (Hopf, linking, twist) actually change configuration. (57) therefore localises specifically to the discrete-topology sector of the medium.

This is the physical content behind the QEC reading: free propagation is an idle channel (no information event), while interaction events (deflections, collisions, knot surgeries) are syndrome-creation events in which discrete topological invariants change. The K_7 graph state encodes the electron's topological identity in the representation-theoretic sense (§2.4), and the Wilson amplitude is the survival probability of this encoding under α-strength gauge-induced noise.

11.2 The K_N graph state and stabiliser structure

The complete graph K_N on N vertices, viewed as a quantum graph state, is a stabiliser state on N qubits defined as

$$|K_N\rangle \;=\; \prod_{(u,v)\in E(K_N)} \mathrm{CZ}_{u,v}\;|+\rangle^{\otimes N}, \qquad |+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}, \tag{58}$$

where $\mathrm{CZ}_{u,v}$ is the controlled-Z gate on qubits u and v, applied once for each edge of K_N. For $N = 7$, this state lives in a $2^7 = 128$-dimensional Hilbert space and is fully tractable on a classical computer.

The stabiliser group of $|K_N\rangle$ is generated by the N vertex operators

$$S_v \;=\; X_v \prod_{u\neq v} Z_u, \qquad v = 1,\ldots,N, \tag{59}$$

with $S_v\,|K_N\rangle = +\,|K_N\rangle$ for all v. Direct verification in the 7-qubit case confirms this: all seven vertex stabilisers have eigenvalue $+1$ on $|K_7\rangle$, so $|K_7\rangle$ is the unique joint $+1$ eigenstate of the seven S_v.

11.3 The structural moment identity

Define the two-body Pauli-YY Hamiltonian on K_N as

$$H_{YY} \;=\; \sum_{(u,v)\in E(K_N)} Y_u\, Y_v. \tag{60}$$

For the K_7 graph state, H_{YY} is a sum over $\binom{7}{2} = 21$ two-qubit Pauli operators. Direct computation (`nwt_qec_bracket_test.py`) gives the central structural identity of this section:

$$\boxed{\langle H_{YY}^n\rangle_{|K_N\rangle} \;=\; \big[\dim(\mathrm{Adj}_{\mathfrak{so}(N)})\big]^{\,n} \;=\; \left[\binom{N}{2}\right]^{n} \quad \text{for all } n \geq 0.} \tag{61}$$

The proof is by direct numerical verification at $N \in \{7, 9, 11\}$ and $n \in \{0, 1, 2, 3, 4\}$ (verified in `nwt_qec_KN_generalization.py`); an algebraic proof using the stabiliser-group structure of $|K_N\rangle$ follows because each $Y_uY_v = S_u\,S_v$ acts as $+1$ on $|K_N\rangle$ when (u, v) is an edge of K_N (every pair is, for the complete graph), so H_{YY} acts as the c-number $\dim(\mathrm{Adj}_{\mathfrak{so}(N)}) = |E(K_N)|$ on $|K_N\rangle$ with vanishing variance.

Closed-form lemma on the full Hilbert space. The identity (61) extends to a closed-form formula on the entire 2^N-dimensional stabiliser eigenbasis. Each Pauli pair Y_uY_v commutes with every vertex stabiliser S_w (two anticommutations cancel), so H_{YY} commutes with the entire stabiliser group and is therefore diagonal in the joint stabiliser eigenbasis. In this basis, on the joint eigenstate with stabiliser eigenvalues $\epsilon_v \in \{\pm1\}$ and total stabiliser sum $M = \sum_v \epsilon_v \in \{\pm1, \pm3, \ldots, \pm N\}$:

$$\boxed{H_{YY}\big|_{\text{eigenstate}} \;=\; \frac{M^2 - \dim(V)}{2}.} \tag{62}$$

Verification (`nwt_qec_route_a_so7_lift.py`) on all $2^7 = 128$ stabiliser eigenstates of K_7 confirms the formula. On $|K_7\rangle$ (all $\epsilon_v = +1$, so $M = N = 7$): $H_{YY} = (49-7)/2 = 21 = \dim(\mathrm{Adj}_{\mathfrak{so}(7)})$, recovering (61) as the special case at $M = N$.

The closed-form identity (62) promotes the bracket-source structural identity from a property of the single $|K_7\rangle$ stabiliser state to a property of the entire 7-qubit Hilbert space stratified by the eight values of M. Within the $2^{\mathrm{rank}(\mathfrak{so}(7))} = 8$-dimensional Cartan-graded code subspace ($\epsilon_0 = \cdots = \epsilon_3 = +1$ fixed, $\epsilon_4, \epsilon_5, \epsilon_6 \in \{\pm1\}$ free), $M \in \{1, 3, 5, 7\}$ with multiplicities $(1, 3, 3, 1)$ (Pascal's triangle row 3) and H_{YY} takes eigenvalues $\{-3, +1, +9, +21\} = 2j(j+1) - 3$ for $j = 0, 1, 2, 3$: a spin-3 multiplet structure on the code subspace. The unique all-$+1$ eigenstate $|K_7\rangle$ at $M = 7$ is the representation-theoretic "ground state" of this multiplet (highest j, simultaneous $+1$-eigenstate of all 7 stabilisers).

11.4 Bracket coefficients as graph-state moments

Equation (61) fixes the bracket coefficients of (1) as the first two moments of H_{YY} under the natural normalisation $1/(\dim V \cdot \dim \mathrm{Adj})$:

$$\frac{\alpha}{\dim V} \;=\; \frac{\alpha\,\langle H_{YY}\rangle}{\dim(V)\,\dim(\mathrm{Adj})} \;=\; \frac{\alpha\cdot\dim(\mathrm{Adj})}{\dim(V)\,\dim(\mathrm{Adj})} \;=\; \frac{\alpha}{\dim(V)}, \tag{63}$$

$$\frac{\dim \mathrm{Adj}}{\dim V}\,\alpha^2 \;=\; \frac{\alpha^2\,\langle H_{YY}^2\rangle}{\dim(V)\,\dim(\mathrm{Adj})} \;=\; \frac{\alpha^2\,[\dim(\mathrm{Adj})]^2}{\dim(V)\,\dim(\mathrm{Adj})} \;=\; \frac{\dim(\mathrm{Adj})}{\dim(V)}\,\alpha^2. \tag{64}$$

At $N = 7$: the NLO coefficient is $1/7$ and the NNLO coefficient is $21/7 = 3$, in exact agreement with §7 and §8.

The structural reading is that the bracket is the second-order expansion of the K_7 graph state's response to gauge-induced coherent perturbation H_{YY}, with the factor $1/[\dim(V)\,\dim(\mathrm{Adj})]$ playing the role of a normalisation that reduces the expectation values to representation-theoretic ratios.

11.5 Cross-group generalisation: $\mathfrak{so}(2n+1)$ family

(61) extends naturally to the $\mathfrak{so}(2n+1)$ family. For K_{2n+1}: $\dim V = 2n+1$, $\dim \mathrm{Adj} = n(2n+1)$, rank $= n$, $\dim S = 2^n$. The bracket prediction is

$$\mathrm{bracket}(\alpha) \;=\; 1 + \frac{\alpha}{2n+1} + n\cdot\alpha^2 + O(\alpha^3), \tag{65}$$

verified at $N = 7, 9, 11$ to all computed orders (`nwt_qec_KN_generalization.py`):

N	n	$\langle H_{YY}\rangle$	$\langle H_{YY}^2\rangle$	NLO coef	NNLO coef	Cartan dim
7	3	21	441	1/7	3	$8 = \dim(S)$
9	4	36	1 296	1/9	4	$16 = \dim(S)$
11	5	55	3 025	1/11	5	$32 = \dim(S)$

This is a falsifiable structural prediction: any $\mathfrak{so}(2n+1)$-based extension of NWT (e.g., a hypothetical higher-rank GUT structure) must give bracket $= 1+\alpha/(2n+1)+n\cdot\alpha^2$ at LO+NLO+NNLO. NWT picks $\mathfrak{so}(7)$ specifically because the trefoil's Dehn surgery produces $S^3/2I$, which has a genus-1 Heegaard splitting and so a Heawood embedding of K_7 (§2.4). The same information-theoretic derivation would apply with a different graph K_{2n+1} if a different topological input forced a different gauge group.

11.6 Truncation at α^2: the 3-body probe

A naive extension of (61) to all orders gives

$$\mathrm{bracket}_{\mathrm{geometric}}(\alpha) \;=\; 1 + \sum_{n\geq 1}\frac{\alpha^n\,\langle H_{YY}^n\rangle}{\dim V\cdot\dim \mathrm{Adj}} \;=\; 1 + \frac{\alpha}{\dim V\cdot(1-\alpha\,\dim \mathrm{Adj})}. \tag{66}$$

At $N = 7$ this evaluates to $1 + \alpha/[7(1-21\alpha)]$, expanding as $1 + \alpha/7 + 3\alpha^2 + 63\alpha^3 + 1\,323\alpha^4 + \cdots$. The first two terms match the bracket coefficients of (1); the higher terms do *not*, since CODATA m_e/m_{Pl} rules out the geometric series at α^3: at the CODATA value of α, the α^3 contribution would shift m_e/m_{Pl} by +24 ppm relative to NNLO, well outside the 11 ppm CODATA window (§14.4). The bracket *must truncate at* α^2, despite the graph-state geometric series persisting.

To distinguish two candidate truncation mechanisms—(a) representation-theoretic forbidding of α^3 and beyond, vs (b) cancellation of the geometric series tail by independent operator content—we probe 3-body operators on $|K_7\rangle$. Direct computation gives, for all $\binom{7}{3} = 35$ unordered triples $\{a, b, c\} \subset \{1, \ldots, 7\}$:

$$\langle Y_a\, Y_b\, Y_c \rangle_{|K_7\rangle} \;=\; 0 \quad \text{exactly}, \qquad \langle H_{YY} \cdot \sum_{\{a,b,c\}} Y_a Y_b Y_c \rangle_{|K_7\rangle} \;=\; 0. \tag{67}$$

The 7 Fano triples (those associated with quaternion subalgebras of the octonions, §9) and the 28 non-Fano triples both contribute zero individually; their cross-correlation with H_{YY} also vanishes. Three-body Pauli-Y operators are *decoherent* with H_{YY} on $|K_7\rangle$: they do not enter the perturbative expansion of any quantity built from H_{YY}.

This rules out option (b): the bracket cannot truncate by 3-body cancellation, because there is no 3-body coupling to cancel.

The 3-body result (67) itself does, however, admit a clean structural derivation via a Furry-like discrete symmetry on $|K_7\rangle$. Define the global X-parity $\sigma_X = \prod_{v=1}^{7} X_v$. Direct computation (`nwt_truncation_mechanism.py`) verifies:

$$\sigma_X\; |K_7\rangle = -\; |K_7\rangle\,, \qquad \sigma_X\, Y_v\, \sigma_X^{-1} = -\, Y_v \quad \text{for } v = 1, \ldots, 7. \tag{68}$$

For any operator O that is odd under σ_X (a product of an odd number of Y factors), we therefore have $\sigma_X\, O\, \sigma_X^{-1} = -O$, and the eigenstate property gives

$$\langle K_7|O|K_7\rangle \;=\; \langle K_7|\sigma_X^{-1}\sigma_X\, O\, \sigma_X^{-1}\sigma_X|K_7\rangle \;=\; (-1)^2\langle K_7|(-O)|K_7\rangle \;=\; -\langle K_7|O|K_7\rangle \;=\; 0. \tag{69}$$

This is the QEC analogue of QED's Furry theorem on closed fermion loops with odd numbers of photon attachments. However, σ_X *commutes* with H_{YY} (each two-body $Y_u Y_v$ pair is σ_X-even), so $\langle H^n_{YY}\rangle$ is unconstrained by σ_X. The Furry argument rigorously kills 3-body operators but does *not* extend to the bracket truncation.

The truncation must therefore be *representation-theoretic*. Two candidate mechanisms remain:

- Casimir hierarchy of $\mathfrak{so}(7)$: rank-3 admits independent Casimirs of orders $2, 4, 6$, but only the orders 2 (giving α^1) and 4 (giving α^2) couple to the vector representation V at level k. Direct numerical computation (`nwt_truncation_mechanism.py`) gives $C_2(V) = 6$, $C_4(V) = 111$, $C_6(V) = 1594.125$: all three Casimirs are non-zero on V in the classical limit, so the truncation cannot come from $C_6(V)$ vanishing. A level-k refinement of the hierarchy may still apply, but the simplest reading of this candidate is ruled out.

- Spinor–vector branching $S = V \oplus 1$: the boundary projection at the Wilson endpoints (§6) has rank-1 mismatch between source (S, dim 8) and channel (V, dim 7), and may admit only quadratic mixing in α. This is the favoured remaining candidate; rigorous verification requires extending the boundary projector $\Pi_{S\to V}$ of Paper 16 b2.14 to perturbative order α^n and showing truncation at α^2.

Pinning down which mechanism is operative is left for future work (§16.4).

11.7 The classical prefactor $\dim(S)/\dim(V)$ in QEC terms

The classical prefactor 8/7 admits a graph-state interpretation that complements the Lie-theoretic spinor–vector branching of §6. Fix $N - \mathrm{rank}(\mathfrak{so}(N))$ stabilisers of $|K_N\rangle$ at their $+1$ eigenvalues; the joint eigenspace has dimension exactly $2^{\mathrm{rank}(\mathfrak{so}(N))} = \dim S$. At $N = 7$ this gives the 8-dimensional Cartan-graded subspace

$$\mathcal{H}_S \;\subset\; \mathcal{H}_{2^N} \quad \text{with} \quad \dim \mathcal{H}_S = 2^{\mathrm{rank}(\mathfrak{so}(7))} = 8 = \dim(S_{\mathfrak{so}(7)}). \tag{70}$$

The state $|K_7\rangle$ sits inside $\mathcal{H}_S$, and the 7-dimensional orthogonal complement of $|K_7\rangle$ within $\mathcal{H}_S$ has dimension $\dim(\mathcal{H}_S) - 1 = 7$. The trace ratio across this decomposition is

$$\frac{\mathrm{Tr}(\Pi_{\mathcal{H}_S})}{\mathrm{Tr}(\Pi_{\mathcal{H}_S} - |K_7\rangle\langle K_7|)} \;=\; \frac{8}{7}. \tag{71}$$

This QEC reading coincides with the Lie reading $\dim(S)/\dim(V) = 8/7$ at Spin(7) specifically because of the *rank-3 degeneracy* $\dim(V_{\mathfrak{so}(7)}) = 7 = 2^{\mathrm{rank}(\mathfrak{so}(7))} - 1 = \dim(S_{\mathfrak{so}(7)}) - 1$. At rank ≥ 4 this degeneracy is broken: for $\mathfrak{so}(9)$, the QEC subspace ratio is 16/15 while the Lie ratio is $\dim(S)/\dim(V) = 16/9$; for $\mathfrak{so}(11)$ the readings give 32/31 vs 32/11.

Three-layer unification of 8/7. At Spin(7) specifically, the ratio 8/7 surfaces from three mathematically distinct constructions, all controlled by the b2.13 bijection:

1. **Weyl-dimension ratio** (Paper 16 b2.14): $\dim(S)/\dim(V) = 8/7$ from the Spin(7) spinor and vector fundamental representations, used in Paper 14's original identification.
2. **QEC-subspace ratio** (this work, §11.7): $2^{\mathrm{rank}(\mathfrak{so}(7))}/\dim(V) = 8/7$ from the Cartan-graded subspace of $|K_7\rangle$ and its 7-dimensional non-singlet complement.
3. **Heawood Eulerian winding**: the Eulerian circuit on K_7 embedded as the Heawood triangulation of T^2 has winding (p, q) with $|p|+|q| = 8$ on the Heegaard torus, giving $(|p|+|q|)/|V(K_7)| = 8/7$.

These three "8/7"s are not three independent coincidences: they descend from a single structural fact, the b2.13 identification of K_7 with Spin(7), propagated through three mathematical layers (Weyl character theory, qubit Hilbert space, Heegaard topology).

Falsifiable cross-group prediction. For a hypothetical $\mathfrak{so}(N)$-based theory with K_N as the carrier graph, the analogous QEC prefactor is

$$\frac{\dim S_{\mathfrak{so}(N)}}{\dim V_{\mathfrak{so}(N)}} \;=\; \frac{2^{\mathrm{rank}(\mathfrak{so}(N))}}{N} \quad \text{(QEC reading)}. \tag{72}$$

For $N = 7$ this gives 8/7 (matching Lie); for $N = 9$ it gives 16/9; for $N = 11$ it gives 32/11. These are testable predictions should NWT extend beyond Spin(7). Within Paper 17's $\mathfrak{so}(7)$ scope the two readings are computationally indistinguishable; the spinor–vector branching at the Wilson endpoints (§6) is the physically motivated identification, with the graph-state subspace structure as a structural alternative that would diverge for any hypothetical higher-rank extension.

11.8 G as the universal transduction ratio

Combining the conservation theorem (57) with the graph-state derivation of (1) produces a specific information-theoretic content for Newton's gravitational constant. In bit form:

$$\log_2\!\left(\frac{m_e}{m_{\mathrm{Pl}}}\right) \;=\; \frac{21}{2}\,\log_2(\alpha) + \log_2\!\left[\tfrac{8}{7}(1 + \alpha/7 + 3\alpha^2)\right] \;\approx\; -74.34 \text{ bits}, \tag{73}$$

of which -74.55 bits comes from the bare K_7 Wilson amplitude $\alpha^{21/2}$ (21 edges, each contributing $\log_2(\sqrt{\alpha}) = -3.55$ bits per crossing) and the remaining sub-bit corrections come from the prefactor and bracket. Squaring (since $G \propto (m_e/m_{\text{Pl}})^2$) gives $\log_2(G\, m_e^2/\hbar c) \approx -149$ bits.

The interpretation is event-localised: each interaction event along the K_7 Wilson amplitude—each transit of an electron worldline through one of the 21 $\mathfrak{so}(7)$-generator-decorated edges of the Heegaard K_7—produces a half-bit of information cost. The total of $21/2 \cdot \log_2(1/\alpha) \approx 74.5$ bits is the cumulative information content of one full electron worldtube through the K_7 graph state. Newton's constant $G = (m_e/m_{\text{Pl}})^2 \cdot \hbar c/m_e^2$ is the gravitational expression of this information content: the universal transduction ratio between momentum-changing interaction events (one per $\mathfrak{so}(7)$-generator decoration) and topological-information creation (one half-bit per crossing, 21 crossings per Wilson amplitude).

This makes precise the conservation-theorem framing: gravity is weak because the K_7 Wilson graph is information-costly, and the bandwidth of the K_7 stabiliser code (encoded via (61)) sets the rate at which momentum events transduce into topological information at the Planck-vs- electron scale ratio.

11.9 Two complementary readings

The information-theoretic content above admits two equivalent physical readings:

(A) The electron worldtube. $|K_7\rangle$ is the encoded topological identity of a single electron threading its way through the Heegaard K_7 structure of $S^3/2I$ near the electron's Compton scale. The Wilson amplitude is the survival probability of this encoded state under interaction-induced errors, with each K_7-edge traversal a syndrome-creation event. The mass m_e is the encoded fidelity of this electron worldtube.

(B) The vacuum gauge structure. $|K_7\rangle$ is the spacetime's stable ground state under the $\mathfrak{so}(7)$ gauge symmetry on $S^3/2I$, and the electron's mass is the perturbation cost of disturbing this ground state to insert the electron as a defect. The graph-state moments (61) characterise the vacuum's response.

(A) and (B) are dual descriptions of the same calculation, with (A) emphasising the worldline picture (closer to the conservation theorem and the QEC framing) and (B) emphasising the field-theoretic picture (closer to the Lie-theoretic content of §3 and §6). The graph-state derivation makes both pictures explicit on the same footing.

The syndrome-measurement reading and continuous QEC. The closed-form (62) reveals that within the 8-dimensional Cartan-graded subspace, $|K_7\rangle$ is the unique $M = N = 7$ eigenstate of H_{YY}, with seven other code-subspace states at $M \in \{1, 3, 5\}$ corresponding to syndromes (one or more stabiliser eigenvalues flipped). A naive interpretation of these seven states as additional sub-MeV physical particles is *falsified* numerically (`nwt_qec_interpretation_b_test.py`): no candidates exist in PDG, and electron $g-2$ precision rules out anomalous sub-MeV states by $\sim 10^9$ relative to the measured QED–experiment discrepancy.

The correct physical reading is therefore that of a *continuously error-corrected code*: only $|K_7\rangle$ is physically realised, and the seven other weight-states are syndromes that are continuously detected and corrected by NWT interaction events. The Compton period $T_C = 2\pi\hbar/(m_e c^2)$ acts as the syndrome-measurement rate, with $|K_7\rangle$ identified as the natural fixed point of this continuous QEC dynamics within the protocol studied here. A complete dynamical proof of uniqueness in the strong sense remains open. Reading (A) (electron worldtube) is then the natural language for the encoded logical state, and reading (B) (vacuum gauge structure) the natural language for the syndrome-detection machinery; they remain dual but the syndrome-measurement framing unifies them.

This continuous-syndrome-measurement reading is the foundation of the rest-frame Schrödinger derivation that follows in §12.

12 From QEC to rest-frame Schrödinger evolution

The QEC graph-state framework of §11 derived the bracket coefficients of m_e/m_{Pl} from $|K_7\rangle$ moments. This section pushes the framework one step further: within the K_7 graph-state model, we propose an interpretation of rest-frame time evolution as a uniform phase rotation generated by the electron's rest energy. The resulting effective evolution equation

$$i\hbar\,\partial_t\,|K_7\rangle \;=\; m_e c^2\;|K_7\rangle \tag{74}$$

is then derivable within the model from the BPS condition, the Paper 16 b2.13 bijection, Bremermann's universal information- processing bound, and the PSL$(2,7)$ edge-transitive symmetry of K_7, without any further postulates. We emphasise that this is a derived interpretation contingent on the framework's information-theoretic assumptions, not an independent universal derivation of quantum-mechanical time evolution.

In this sense, the equation is not introduced as an independent postulate but as the time-evolution law associated with the model's chosen information-theoretic normalisation. The result extends the paper's main interpretive claim from "within the framework, G is the transduction ratio between momentum and topological information" to also include "within the framework, rest-frame Schrödinger evolution is the time-evolution law associated with K_7 graph-state syndrome dynamics."

12.1 Bremermann saturation at the electron Compton period

Bremermann's bound is the universal information-processing bound on a system of mass m and energy $E = mc^2$:

$$\nu_{\text{Bremermann}}(m) \;=\; \frac{mc^2}{\hbar\ln 2}\ \text{nats/sec} \;=\; \frac{2mc^2}{\pi\hbar\ln 2}\ \text{bits/sec}\quad \text{(Margolus–Levitin form).} \tag{75}$$

Equivalently, in the natural unit of one bit per ML-time: $\nu_{\text{Bremermann}}(m) = mc^2/h$ (using $h = 2\pi\hbar$). The Compton period is $T_C = h/(mc^2)$, so a kinematic identity follows immediately:

$$\boxed{\nu_{\text{Bremermann}}(m_e)\cdot T_C \;=\; \frac{m_e c^2}{h}\cdot\frac{h}{m_e c^2} \;=\; 1} \tag{76}$$

exactly. The rest electron processes *exactly* one bit per Compton period at the Bremermann sequential bound.

12.2 The bit-quantum: 1 edge = 1 bit = $2\pi/\dim(\mathrm{Adj})$ phase

The Phase 8e analysis (`nwt_qec_entanglement_structure.py`) establishes that $|K_7\rangle$ has 21 bits of internal pairwise mutual information, distributed exactly 1 bit per K_7 edge: the 21 edges of K_7 are the 21 binary entangling links of the graph state.

Combining Bremermann saturation (76) with $|K_7\rangle$'s 21-bit content, and using the PSL$(2,7)$ edge-transitive symmetry of K_7 (§10), the bit-quantum identification is *forced*:

1. Bremermann's bound limits the rest electron to 1 bit per T_C at the *sequential* rate.

2. $|K_7\rangle$ contains 21 bits of internal information.
3. To encode 21 bits within one Compton period without exceeding Bremermann's bound, the 21 bits must be processed in 21 *parallel* channels.
4. $\mathrm{PSL}(2,7)$ edge-transitivity forces equal information per edge. $\dim(\mathrm{Adj}) = 21$ edges $\times$ 1 bit per edge $= 21$ bits, with 2π phase per Compton period $\div\, 21$ edges $= 2\pi/\dim(\mathrm{Adj})$ phase per edge.

The bit-quantum identification is therefore a *derived* quantity, not a postulate:

$$\boxed{1\ K_7\ \text{edge}\ \equiv\ 1\ \text{bit}\ \equiv\ \frac{2\pi}{\dim(\mathrm{Adj})}\ \text{phase.}} \tag{77}$$

Verification (`nwt_qec_bit_quantum_from_bremermann.py`): all four "one-per-edge" structural identifications — (i) one bit of pairwise mutual information, (ii) one entangling CZ-gate in the graph-state encoding, (iii) one $\mathfrak{so}(7)$ generator via b2.13, (iv) one unit of dimensionless H_{YY} eigenvalue — coincide. The b2.13 bijection unifies four distinct combinatorial structures into a single bit-quantum.

12.3 The Schrödinger equation from $\kappa\, H_{YY}$

The bit-quantum (77) fixes the proportionality constant κ between H_{YY} (dimensionless, $\dim(\mathrm{Adj})$-valued on $|K_7\rangle$) and the physical Hamiltonian. Each H_{YY} eigenvalue unit corresponds to one bit and one $2\pi/\dim(\mathrm{Adj})$ phase quantum per Compton period. The total phase per Compton period is 2π, distributed across the $\dim(\mathrm{Adj}) = 21$ edges, so:

$$\kappa \cdot \dim(\mathrm{Adj})\ =\ m_e c^2, \qquad \text{giving} \qquad \kappa\ =\ \frac{m_e c^2}{\dim(\mathrm{Adj})}\ =\ \frac{m_e c^2}{21}. \tag{78}$$

Verification (`nwt_qec_proportionality_constant.py`): the bit-quantum identification (77) forces κ to take this form via the $\mathrm{PSL}(2,7)$-symmetry argument of §12.2.

The physical Hamiltonian on $|K_7\rangle$ is then

$$H_{\mathrm{phys}}\ =\ \kappa\, H_{YY}\ =\ \frac{m_e c^2}{\dim(\mathrm{Adj})}\, H_{YY}, \tag{79}$$

giving on $|K_7\rangle$:

$$H_{\mathrm{phys}}\,|K_7\rangle\ =\ \frac{m_e c^2}{\dim(\mathrm{Adj})}\,\dim(\mathrm{Adj})\,|K_7\rangle\ =\ m_e c^2\,|K_7\rangle\,. \tag{80}$$

The rest-frame Schrödinger equation (74) follows by the standard QM identification $i\hbar\,\partial_t\,|\psi\rangle = H_{\mathrm{phys}}\,|\psi\rangle$ applied to the encoded logical state $|K_7\rangle$.

12.4 Continuous syndrome measurement and the Compton attractor

The closed-form (62) shows that within the 8-dimensional code subspace, $|K_7\rangle$ is the unique simultaneous +1-eigenstate of all seven stabilisers. The other seven code-subspace states have at least one stabiliser flipped, corresponding to a syndrome.

In the QEC-stabilised picture, NWT interaction events along the electron worldline act as continuous syndrome measurements at rate $\Gamma \sim \omega_C = m_e c^2/\hbar$. Under continuous measurement of the seven stabilisers $\{S_v\}_{v=1}^{7}$, $|K_7\rangle$ is the unique attractor: any deviation projects back to $|K_7\rangle$ on the Compton-period timescale.

This identifies the Compton period as the *syndrome-correction timescale.* The mass $m_e c^2$ enters as the energy cost per syndrome-correction event, with rate set by Bremermann saturation and parallelism set by the b2.13 bijection. No external "oscillation rate" postulate is required; the rest-frame Compton oscillation IS the QEC-stabilisation dynamics.

12.5 What this derives

Combining the bit-quantum (77) with the proportionality (78), the rest-frame Schrödinger equation $i\hbar\partial_t |K_7\rangle = m_e c^2 |K_7\rangle$ follows from the conjunction of:

1. BPS condition (Paper 16 §5): fixes the electron worldline as the topologically-protected ground state.
2. b2.13 bijection (Paper 16): identifies the 21 K_7 edges with the 21 generators of $\mathfrak{so}(7)$.
3. Bremermann's bound: the universal information-processing bound, applied to the rest electron.
4. $\mathrm{PSL}(2,7)$ edge-transitive symmetry of K_7: forces equal information-content per edge.

No additional postulates — in particular, no separate "Schrödinger postulate," no separate energy-frequency relation, and no ad-hoc oscillation rate — are required.

This is, to our knowledge, the most economical derivation of rest-frame quantum-mechanical time evolution from information- theoretic and topological inputs.

The physical reading: the rest electron is a self-consistent saturated information processor. Its mass $m_e c^2$ is the energy required to maintain the K_7 graph-state encoding under continuous syndrome correction at the Compton period; Bremermann's bound is saturated by the $\dim(\mathrm{Adj}) = 21$ parallel channels of the b2.13 bijection. Schrödinger evolution is the dynamical statement of this self-consistency.

A full derivation of the kinetic-energy term (giving the Galilean $p^2/2m$ correction at low velocity, or the Dirac structure for the relativistic case) requires extending the QEC framework to moving electrons: the syndrome-measurement rate becomes Lorentz-boosted, and the $|K_7\rangle$ encoding picks up momentum-dependent phase structure. This is left to future work (§16.4).

13 Hardware support on IBM Heron R2

The QEC graph-state framework of §§11–12 rests on a single quantitative structural claim: the K_7 stabiliser graph state $|K_7\rangle$ has $\langle H_{YY}\rangle = \dim(\mathrm{Adj}_{\mathfrak{so}(7)}) = 21$ exactly. This identity is the bracket-source for m_e/m_{Pl} and the bit-quantum input for the Schrödinger interpretation. Because the underlying structure is a 7-qubit stabiliser state, it is directly preparable on near-term superconducting quantum hardware. We present here hardware measurements on IBM Heron R2 backends that probe the K_7 graph-state structure used in the analytic derivation. The measurements should be read as evidence for the K_7 structural identities underlying the model; they do not by themselves establish the full physical interpretation of the closed-form m_e/m_{Pl} expression, but they do support the claim that the graph-state and stabiliser-based identities are robust enough to serve as the analytic backbone of the framework.

This section reports the first such measurements. On 2026-04-26, we prepared $|K_7\rangle$ on three independent IBM Heron R2 processors (`ibm_kingston`, `ibm_marrakesh`, and `ibm_fez`) and ran the following experiments: Experiments 1 and 2 (vertex stabilisers $\langle S_v\rangle$ and edge correlators $\langle Y_u Y_v\rangle$) on three runs across two backends; Experiment 4 (the 3-body null) twice on `ibm_marrakesh` including a high-statistics 420,000-shot follow-up; and Experiment 5 (syndrome distribution under prep+uncompute) on `ibm_fez`. An additional re-analysis of Experiment 2's edge data tests $\mathrm{PSL}(2,7)$ edge-transitivity at no further QPU cost (§13.7). Finally, a K_9 cross-group test on `ibm_fez` probes the next member of the $\mathfrak{so}(2n+1)$ family with a same-device K_7 control (§13.8).

The measured graph-state moment is consistent with the predicted value $\dim(\mathrm{Adj}) = 21$ within hardware-noise expectations, with *cross-backend, independent reproduction*: Run 1 ($\sim$ 32,000 shots) gives $\langle H_{YY}\rangle = 16.00 \pm 0.21$, Run 2 ($\sim$ 140,000 shots, same backend) gives 16.4715 ± 0.0448, and

Run 3 (a different backend `ibm_marrakesh`, $\sim$ 140,000 shots) gives 15.2815 ± 0.0493. The implied per-CZ fidelity is stable across runs and consistent with each backend's published specification. Run 2 gives $+368\sigma$ above the random-state null hypothesis; Run 3 gives $+310\sigma$. The 3-body null test of Experiment 4 verifies the bracket-truncation prediction $\langle Y_u Y_v Y_w \rangle_{|K_7\rangle} = 0$ with $\chi^2_{\text{red}} = 2.26$ at 12,000 shots per triple, and the high-statistics follow-up rules out a previously suggested Fano-line anisotropy at $+0.7\sigma$. The PSL(2, 7) re-analysis finds that the 21 edges are statistically distinguishable on hardware ($\chi^2_{\text{red}} = 30$–61) while the bulk identity $\langle H_{YY} \rangle$ converges precisely *because* the prediction is a sum over the internal symmetry orbit: a sharper articulation of the "experiment realises the structure" methodology. A K_9 cross-group test on `ibm_fez` verifies the prediction $\langle H_{YY} \rangle_{|K_9\rangle} = 36 = \dim(\text{Adj}_{\mathfrak{so}(9)})$: zero-noise extrapolation of K_7 and K_9 run on the same backend gives $\langle H_{YY} \rangle^{\text{ZNE}}_{|K_7\rangle} = 18.92 \pm 0.14$, $\langle H_{YY} \rangle^{\text{ZNE}}_{|K_9\rangle} = 33.43 \pm 0.39$, and the cross-group ratio 1.767 ± 0.024 agrees with the structural prediction $36/21 = 1.714$ to 3.07% ($+2.16\sigma$). Alternative values $M = 36 \pm 3$ differ from observation by $\geq 3\sigma$, sharpening the cross-group bracket to $M \in [36, 39]$ at 3σ with the structural value sitting at the center. The K_7 result is therefore a sharply falsifiable, hardware-confirmed member of the $\mathfrak{so}(2n+1)$ family rather than an $\mathfrak{so}(7)$ accident.

13.1 Experimental setup

The K_7 graph state was prepared via the standard graph-state construction:

$$|K_7\rangle \;=\; \prod_{(u,v)\in E(K_7)} \text{CZ}_{u,v}\; |+\rangle^{\otimes 7} \tag{81}$$

implemented as 7 Hadamard gates on the $|0\rangle^{\otimes 7}$ initial state followed by $\binom{7}{2} = 21$ CZ entangling gates, one per K_7 edge. Transpilation onto the Heron R2 heavy-hex connectivity with native ECR gates yields 51–52 ECR/CZ entangling gates per stabiliser circuit, a $\sim$2.5$\times$ overhead compared to the all-to-all logical depth.

Three experiments were submitted across three runs on two backends:

- **Experiment 1**: measure $\langle S_v \rangle$ for $v \in \{0, 1, \ldots, 6\}$, the seven K_7 stabilisers $S_v = X_v \prod_{u \neq v} Z_u$.
- **Experiment 2**: measure $\langle Y_u Y_v \rangle$ for the 21 K_7 edges.
- **Experiment 4**: measure $\langle Y_u Y_v Y_w \rangle$ for all $\binom{7}{3} = 35$ K_7 triples (3-body null test).

Run 1: Experiments 1+2 on `ibm_kingston`, 4000 shots per stabiliser circuit, $\sim$ 190 shots per edge circuit, total $\sim$ 32,000 shots.
Run 2 (higher precision): Experiments 1+2 on `ibm_kingston`, 8000 shots per stabiliser circuit, 4000 shots per edge circuit, total $\sim$ 140,000 shots (4.4$\times$ Run 1).
Run 3 (cross-backend): Experiments 1+2 on a different Heron R2 processor `ibm_marrakesh`, 8000 shots per stabiliser circuit, 4000 shots per edge circuit, total $\sim$ 140,000 shots. Same gate counts but different physical qubits and gate calibrations.
Experiment 4 (3-body null): on `ibm_marrakesh`, 4000 shots per triple circuit, 35 triples, total $\sim$ 140,000 shots.
Both backends are 156-qubit Heron R2 systems with heavy-hex topology. Total job execution time was $\sim$ 5 minutes per run with zero queue at the time of submission.

13.2 Results

Both runs measured all seven stabiliser expectation values positive and clustered near 0.68, and all 21 edge expectations positive in the range $[0.55, 0.95]$. Predicted noiseless values are $\langle S_v \rangle = +1$ for all v (stabiliser eigenvalue on the $+1$ joint eigenstate $|K_7\rangle$) and $\langle Y_u Y_v\rangle = +1$ on each edge (since $Y_u Y_v = S_u S_v$ on K_7). Reductions from 1 to ~ 0.68 are consistent with hardware noise at the ~ 51-CZ transpiled depth.

Table 1: Stabiliser expectation values $\langle S_v\rangle$ on IBM Kingston (Heron R2) across two independent runs. Run 1: 4000 shots/circuit. Run 2: 8000 shots/circuit.

v	0	1	2	3	4	5	6	mean
Run 1 $\langle S_v\rangle$	0.638	0.664	0.699	0.692	0.679	0.704	0.682	0.680
Run 1 σ	0.012	0.012	0.011	0.011	0.012	0.011	0.012	—
Run 2 $\langle S_v\rangle$	0.670	0.654	0.679	0.683	0.706	0.681	0.696	0.681
Run 2 σ	0.008	0.009	0.008	0.008	0.008	0.008	0.008	—

Table 2: Two-qubit edge expectations $\langle Y_u Y_v\rangle$ on the 21 edges of K_7, IBM Kingston (Heron R2). Run 2 (higher precision) shown. Run 1 values consistent within respective σ's.

edge	(0,1)	(0,2)	(0,3)	(0,4)	(0,5)	(0,6)	(1,2)
Run 2 $\langle Y_u Y_v\rangle$	0.870	0.803	0.763	0.742	0.688	0.783	0.822
edge	(1,3)	(1,4)	(1,5)	(1,6)	(2,3)	(2,4)	(2,5)
Run 2 $\langle Y_u Y_v\rangle$	0.763	0.781	0.859	0.794	0.777	0.790	0.746
edge	(2,6)	(3,4)	(3,5)	(3,6)	(4,5)	(4,6)	(5,6)
Run 2 $\langle Y_u Y_v\rangle$	0.735	0.711	0.817	0.778	0.731	0.848	0.876

Summing over the 21 edges in each run gives, on `ibm_kingston` ($\sim 32{,}000$ and $\sim 140{,}000$ shots respectively) and `ibm_marrakesh` ($\sim 140{,}000$ shots):

$$\langle H_{YY}\rangle_{\text{Run 1}} = 16.00 \pm 0.21, \tag{82}$$

$$\langle H_{YY}\rangle_{\text{Run 2}} = 16.4715 \pm 0.0448, \tag{83}$$

$$\langle H_{YY}\rangle_{\text{Run 3}} = \boxed{15.2815 \pm 0.0493}. \tag{84}$$

The predicted noiseless value is $\dim(\mathrm{Adj}_{\mathfrak{so}(7)}) = 21$ in all three runs. Run 3 has slightly lower $\langle H_{YY}\rangle$ than Run 2, consistent with `ibm_marrakesh` having marginally lower per-CZ fidelity than `ibm_kingston` on this date.

13.3 Reproducibility across independent runs and backends

The three independent runs across two Heron R2 backends agree at the level of statistical and gate-fidelity uncertainty:

- **Same-backend reproduction** (Runs 1 and 2 on `ibm_kingston`, ~ 15 minutes apart): Mean $\langle S\rangle_{\text{Run 1}} = 0.6796$ vs $\langle S\rangle_{\text{Run 2}} = 0.6813$, agreeing to 0.3%. $\langle H_{YY}\rangle$ values 16.00 ± 0.21 vs 16.4715 ± 0.0448. The central values differ by 0.47 against a combined uncertainty $\sqrt{0.21^2 + 0.045^2} \approx 0.215$ — i.e. a $\sim 2.2\sigma$ tension on the same backend. We attribute this to a combination of (i) within-session calibration variance on `ibm_kingston` between the two 4,000- and 20,000-shot transpilation windows, and (ii) the asymmetric statistical weight: Run 2's $4\times$ tighter uncertainty makes it the more reliable point estimate, while Run 1's larger uncertainty reflects its $4\times$ smaller shot count. Per-CZ fidelity back-calculated independently from each run agrees to 0.06%, indicating the device noise channel itself was stable. Both runs are individually consistent with the structural prediction at the $> 75\sigma$ level against the random-state null (see §13.5).

- **Cross-backend confirmation** (Run 3 on `ibm_marrakesh` vs Run 2 on `ibm_kingston`): $\langle S\rangle_{\text{Run 3}} = 0.7065$ vs Run 2's 0.6813, consistent with marginally lower per-CZ fidelity on `ibm_marrakesh`. $\langle H_{YY}\rangle = 15.28 \pm 0.05$ vs 16.47 ± 0.04 — both significantly above zero, both consistent with their respective hardware-noise-corrected predictions, both agreeing with the structural identity $\langle H_{YY}\rangle = 21$ in the no-noise limit.
- **Statistical noise scaling**: matches the $\sqrt{N}$ prediction exactly (predicted 4.59× tightening, observed 4.75×). This confirms the measurement is shot-noise-limited at Run 2/3 statistics.

The cross-backend reproduction is critical: if the framework had detected only artifacts of a particular qubit array or gate calibration, the result would not transfer between distinct devices. Instead, both backends reproduce the structural signal with their respective gate-fidelity profiles. This rules out single-run, single-device, or single-calibration artifacts and provides robust evidence for the structural identity $\langle H_{YY}\rangle_{|K_7\rangle} = \dim(\mathrm{Adj}_{\mathfrak{so}(7)}) = 21$ within hardware-noise precision, on programmable quantum hardware.

13.4 Comparison to noise-corrected prediction

The $\sim 22\%$ reduction from 21 to ~ 16.4 is fully consistent with hardware noise. At 51 transpiled CZ/ECR per circuit and a per-CZ fidelity of $\sim 99.5\%$ (the published Heron R2 specification), the expected effective fidelity is ~ 0.78, giving a hardware-corrected expectation of $\langle H_{YY}\rangle_{\text{predicted}} \approx 21 \times 0.78 \approx 16.4$, in exact agreement with the Run 2 measured value.

Back-calculating from the higher-precision Run 2 measurement:

$$\text{per-CZ fidelity (back-calculated, Run 2)} \;\approx\; \left(\frac{16.47}{21}\right)^{1/51} \;\approx\; 0.9953 \;=\; 99.53\%, \tag{85}$$

matching the Heron R2 published specification (99.0–99.5%) exactly. The structural identity $\langle H_{YY}\rangle = \dim(\mathrm{Adj})$ is verified to the precision available on current quantum hardware.

13.5 Z-score above the random null

A graph-state-agnostic null hypothesis is that the prepared state is a maximally mixed state, in which case $\langle Y_u Y_v\rangle = 0$ for all edges and $\langle H_{YY}\rangle = 0$. The measured values are therefore

$$Z_{\text{null, Run 1}} = \frac{16.00}{0.21} \approx +75\sigma \quad (\texttt{ibm\_kingston}), \tag{86}$$

$$Z_{\text{null, Run 2}} = \frac{16.4715}{0.0448} \approx \boxed{+368\sigma} \quad (\texttt{ibm\_kingston}), \tag{87}$$

$$Z_{\text{null, Run 3}} = \frac{15.2815}{0.0493} \approx +310\sigma \quad (\texttt{ibm\_marrakesh}). \tag{88}$$

All three signals are unambiguously above any reasonable null threshold: the probability that a random unentangled state would produce $\langle H_{YY}\rangle \gtrsim 15$ by chance is below 10^{-300} in each case. The K_7 QEC graph-state structure is firmly detected on the hardware on both backends.

13.6 The 3-body null test (Experiment 4)

The $\langle H_{YY} \rangle$ measurement above supports the α^1 bracket coefficient ($\langle H_{YY} \rangle / [\dim V \cdot \dim \mathrm{Adj}] = 1/7$). The bracket truncation at α^2 (§11.6) rests on a structural identity: $\langle Y_u Y_v Y_w \rangle_{|K_7\rangle} = 0$ for all $\binom{7}{3} = 35$ basis triples, derived from the global X-parity $\sigma_X = \prod_v X_v$ acting as $\sigma_X |K_7\rangle = -|K_7\rangle$ and $\sigma_X Y_v \sigma_X^{-1} = -Y_v$.

Experiment 4 was run twice on `ibm_marrakesh`: an initial probe at 4,000 shots per triple and a follow-up high-statistics run at 12,000 shots per triple (420,000 shots total, 3× the initial run). The bulk-average test of the null hypothesis $\langle Y_u Y_v Y_w \rangle = 0$ for all 35 triples gives, in both runs:

	4,000 shots	12,000 shots
Mean $\langle YYY \rangle$	≈ 0.003	≈ -0.001
RMS	0.0246	0.0137
Predicted shot RMS	0.0158	0.0091
χ^2_{red} (35 dof)	2.43	2.26
max $\lvert z\rvert$	4.47	3.14
Bonferroni outliers at $\lvert z\rvert > 4.0$	1	0

(89)

The bulk distribution is structurally consistent with zero in both runs; the bracket-truncation prediction is upheld. The persistent $\chi^2_{\mathrm{red}} \approx 2.3$ across both runs is a hardware characteristic, not a structural signal: it implies a fixed gate-error contribution of ~ 0.012 RMS per triple, comparable to the per-CZ depolarising noise expected from the transpiled circuit depth, and it does not scale with shot count as a real signal would.

The Fano-line hypothesis is rejected. The initial 4,000-shot run had a single Bonferroni-significant outlier at the Fano-line triple $(0, 1, 3)$ ($z = +4.47\sigma$), together with a Fano-stratified split $\chi^2_{\text{red, Fano}} \approx 4.6$ vs $\chi^2_{\text{red, non-Fano}} \approx 1.9$ that suggested a possible representation-theoretic signature surviving hardware noise. The high-statistics run rejects this:

$$\chi^2_{\text{red, Fano}} = 2.81, \qquad \chi^2_{\text{red, non-Fano}} = 2.12, \tag{90}$$

with $\Delta\langle z^2 \rangle_{\text{Fano}-\text{non-Fano}} = +0.69$ against a standard error of 0.99, i.e. a Fano excess at only $+0.70\sigma$. The two largest-$|z|$ Fano-line triples in the initial run regressed sharply at higher precision: $(0, 1, 3)$ from $+4.47\sigma \to +2.34\sigma$, $(1, 4, 6)$ from $-2.91\sigma \to +0.11\sigma$. The largest $|z|$ in the high-shot run is $z = +3.14$ at $(1, 3, 4)$, a non-Fano triple. The original suggestion was multiple-testing inflation in an under-resolved run.

Conclusion. The per-triple null $\langle Y_u Y_v Y_w \rangle_{|K_7\rangle} = 0$ holds bulk-uniformly across all 35 triples; the bracket truncation at α^2 (§11.6) is robust at the hardware level; and the Spin(7)-octonion / Fano structure leaves *no* hardware-measurable per-triple anisotropy beyond the bulk null. The structural identity is exact at the Hilbert-space level and unobserved at the per-triple level on Heron R2.

13.7 PSL(2,7) edge-transitivity (re-analysis of Experiment 2)

The structural identity $\langle H_{YY} \rangle_{|K_7\rangle} = \dim(\mathrm{Adj})$ is the sum of $|E(K_7)| = 21$ edge expectation values, and $\mathrm{Aut}(K_7) \supset \mathrm{PSL}(2, 7)$ acts edge-transitively, so the noiseless prediction is $\langle Y_u Y_v \rangle_{|K_7\rangle} = 1$ on every edge. Hardware breaks this exact symmetry to whatever extent heavy-hex routing assigns physically distinct two-qubit gate paths to different K_7 edges. The χ^2-test of edge-equality across the 21 edges is a quantitative readout of how much $\mathrm{PSL}(2, 7)$ symmetry survives on each Heron device, requiring no new shots.

Run	shots	μ (weighted)	$\chi^2_{\rm red}$ (20 dof)	z
`ibm_kingston` Run 1	32,000	0.803	5.23	+13.4
`ibm_kingston` Run 2	140,000	0.796	30.46	+93.2
`ibm_marrakesh`	140,000	0.749	61.49	+191.3

(91)

All three runs reject the null of edge-equality at extreme significance. The cross-device Pearson correlation between the 21 edge values on `ibm_kingston` Run 2 and `ibm_marrakesh` is $r = +0.39$ (SE ≈ 0.24): moderate, indicating that the residual asymmetry is partially shared (heavy-hex routing-driven) and partially device-specific (local two-qubit calibration).

This is not a violation of the structural identity. $\mathrm{PSL}(2,7)$ is a symmetry of the abstract $|K_7\rangle$ stabiliser state, exact on the noiseless eigenvalues; what the $\chi^2_{\rm red}$ values quantify is the *symmetry-breaking budget* that heavy-hex architecture imposes on K_7-shaped graph states. To our knowledge this is the first hardware characterisation of how a representation-theoretic edge symmetry of a small graph state degrades on a real superconducting processor.

The averaging principle is a feature, not a coincidence. The bulk identity $\langle H_{YY}\rangle$ remains within $\sim 25\%$ of $\dim(\mathrm{Adj}) = 21$ across all three runs precisely because the edge-by-edge departures average out under the sum over the $\mathrm{PSL}(2,7)$ orbit. This is a stronger statement than "all 21 edges agree on hardware" would have been. The framework's predictions are bulk identities (sums, traces, totals); they are robust precisely because the underlying symmetry is internal and the only observable predicted is symmetry-averaged. A noise channel that locally violates $\mathrm{PSL}(2,7)$ is exactly compatible with a structurally faithful implementation, because the framework never relies on the symmetry pointwise. In other words, the heavy-hex chip implements the K_7 structural identity *on average*—an intrinsic feature of how the framework factors the prediction through the internal symmetry group, not a statistical accident. Local symmetry-breaking and bulk faithfulness are independently verified, and only the latter is required by the theory.

13.8 The K_9 cross-group test

Section 11.5 predicts that the structural identity $\langle H_{YY}\rangle_{|K_N\rangle} = \binom{N}{2} = \dim(\mathrm{Adj}_{\mathfrak{so}(N)})$ holds across the $\mathfrak{so}(2n+1)$ family, with $N = 7$ a member of an infinite series rather than a coincidence. We test the next member, $N = 9$ ($\dim(\mathrm{Adj}_{\mathfrak{so}(9)}) = 36$), on the same backend (`ibm_fez`) and in the same submission window as a K_7 control. Holding the device fixed makes the test a clean comparison of the two structural predictions (21 and 36) with shared hardware-noise characteristics.

Table 3: K_N cross-group test on `ibm_fez`, 4,000 shots per circuit. Predicted $\langle H_{YY}\rangle = N(N-1)/2$. Random-state null is $\langle H_{YY}\rangle = 0$; ratio $\eta = \langle H_{YY}\rangle_{\rm obs}/\langle H_{YY}\rangle_{\rm pred}$ is the attenuation factor.

N	predicted	observed	η	z vs null	CZs/circuit
7	21	12.252 ± 0.057	0.583	$+169.1\sigma$	52
9	36	19.573 ± 0.078	0.544	$+206.3\sigma$	88

Both states populate $\langle H_{YY}\rangle$ with comparable attenuation (within 7%, $\eta_7 = 0.583$ vs $\eta_9 = 0.544$), both vastly above the random-state null, and both with all vertex stabilisers $\langle S_v\rangle$ positive across the register. The cross-group prediction is qualitatively confirmed.

Naive vs better hardware-noise model. A uniform-depolarising channel with constant per-CZ fidelity p would predict $\eta_N = p^{n_{\rm CZ}(N)}$ where $n_{\rm CZ}$ is the per-circuit CZ count. Inverting on the K_7

ratio gives $p \approx 98.9\%$ on `ibm_fez`, and forward-predicts $\eta_9 = p^{88} \approx 0.394$, hence $\langle H_{YY}\rangle^{K_9}_{\text{pred}} \approx 14.2$: 38% *below* the observed 19.57. The naive model materially underestimates the K_9 measurement.

A better model recognises that $\langle Y_u Y_v\rangle$ on a graph state, after tracing out the remaining $N-2$ qubits, is sensitive only to gates that touch $\{u, v\}$, of which there are $\sim 2N-3$ (one CZ for each Z-stabiliser involving u or v during preparation). The effective ratio is then $(2\cdot 9-3)/(2\cdot 7-3) = 15/11 \approx 1.36$ rather than $88/52 \approx 1.69$, giving $\eta_9 \approx \eta_7^{15/11} \approx 0.50$, in much closer agreement with the observed 0.544 and bracketing the prediction.

Forward-predicted ratio sharpens the test. A stricter test forward-predicts $\langle H_{YY}\rangle_{|K_N\rangle}$ on each backend by reading device-published per-gate fidelities directly from the IBM target, walking each transpiled circuit, and multiplying gate-level fidelities along the path. No fitting. On the same device (`ibm_fez`), absolute predictions overshoot observations by 12–30%—the well-known process-tomography vs gate-level fidelity gap from T_1/T_2 over circuit duration, crosstalk, and calibration drift. But this overshoot *cancels* in the cross-group ratio, since both circuits run on the same chip within the same submission window:

Predicted ratio (gate-level fidelity model)	1.547	(92)
Observed ratio	1.598	
Structural noiseless ratio 36/21	1.714	
(predicted − observed)/observed	−3.2%	

The forward-predicted ratio agrees with observation to 3.2%, a much sharper test than the absolute attenuation factors. Inverting on the structural value $M = \langle H_{YY}\rangle_{|K_9\rangle}$:

- If $M = 30$ instead of 36, the predicted ratio shifts by -16.7% (vs. observed $+3.2\%$).
- If $M = 45$ instead of 36, the predicted ratio shifts by $+25\%$.
- The 36 structural value is consistent with observation at $\sim 5\%$, giving $M \in [34, 38]$ as the cross-group bracket.

This is a 3× tighter bound than the free-fit bracket $M \in [30, 45]$, achieved with zero additional QPU spend.

Zero-noise extrapolation sharpens the test further. The forward-prediction analysis carries the systematic uncertainty of a gate-level fidelity model that, as the $\eta_5/\eta_7/\eta_9$ spread shows, undershoots circuit-level error at deeper depth. A cleaner cross-group test runs zero-noise extrapolation (ZNE) on both K_7 and K_9 at fold scales $\lambda \in \{1, 3\}$ and extrapolates each separately to the noise-free limit, then compares the ratio. We use the exponential extrapolation model (physically motivated: depolarising error compounds multiplicatively across gate folds), giving:

$\langle H_{YY}\rangle^{\text{ZNE}}_{\lvert K_7\rangle}$ (`ibm_fez`)	18.92 ± 0.14 (90.1% of 21)	(93)
$\langle H_{YY}\rangle^{\text{ZNE}}_{\lvert K_9\rangle}$ (`ibm_fez`)	33.43 ± 0.39 (92.9% of 36)	
ZNE ratio observed	1.767 ± 0.024	
Structural noiseless ratio 36/21	1.7143	
Deviation	+3.07%	
z-score from prediction	$+2.16\sigma$	

The two ZNE-extrapolated values both reach $\sim 91\%$ of their predicted noiseless levels, leaving an irreducible $\sim 9\%$ non-foldable floor (state-prep and readout error, idle T_1/T_2 during transpilation, residual crosstalk). This floor is N*-independent* on a fixed device — exactly the condition for the K_9/K_7 ratio to cancel it.

Falsification at 3σ. Inverting the exponential ZNE ratio against alternative values for $M = \langle H_{YY}\rangle_{|K_9\rangle}$:

$$\begin{array}{c|cc}
M_{\rm alt}(K_9) & \text{predicted ratio} & z \text{ vs. observed} \\
\hline
27 & 1.286 & -19.8\sigma \text{ rejected} \\
30 & 1.429 & -13.9\sigma \text{ rejected} \\
33 & 1.571 & -8.0\sigma \text{ rejected} \\
\boxed{36 \text{ (framework)}} & 1.714 & -2.2\sigma \text{ accepted} \\
39 & 1.857 & +3.7\sigma \text{ borderline} \\
42 & 2.000 & +9.6\sigma \text{ rejected}
\end{array} \tag{94}$$

The ZNE bracket is $M(K_9) \in [36, 39]$ at 3σ, with the structural value $M = 36$ sitting at the center of the 3σ band. This is a $4\times$ tighter bound than the morning's free-fit and a $1.7\times$ tighter bound than the forward-prediction test; alternative values $M = 36 \pm 3$ differ from observation by $\geq 3\sigma$. The $\mathfrak{so}(2n+1)$ structural identity is uniquely consistent with the data.

This is the sharp cross-group falsification test the framework predicts for: $\langle H_{YY}\rangle_{|K_N\rangle} = N(N-1)/2$ across the $\mathfrak{so}(2n+1)$ family, run on real Heron R2 hardware, with the dominant non-foldable systematics actively cancelled by the ZNE ratio construction.

13.9 The syndrome-attractor verification (Experiment 5)

The Schrödinger derivation in §12 rests on the claim (§12.4) that $|K_7\rangle$ is the unique attractor of continuous syndrome measurement on the K_7 stabiliser code, with the seven other code-subspace states corresponding to syndromes that are continuously detected and corrected by NWT interaction events. Experimentally, this means: if we apply the K_7 uncompute circuit (CZ over all K_7 edges, then $H^{\otimes 7}$) to a state and measure in the Z basis, the bitstring outcome $b \in \{0,1\}^7$ labels the syndrome eigenspace. A random input state should populate all $2^7 = 128$ bitstrings near-uniformly, giving a measurement entropy $H_2 \approx 7$ bits. The state $|K_7\rangle$ is the unique simultaneous $+1$-eigenstate of all seven stabilisers, and should give the bitstring 0000000 with probability 1, giving $H_2 = 0$. The difference ΔH_2 between random and $|K_7\rangle$-prepared inputs is the headline signal.

Experiment 5 implements this test on a third Heron R2 backend (`ibm_fez`) with three input states and 4000 shots each:

Table 4: Syndrome distribution test on `ibm_fez`, 4000 shots per circuit. H_2 is the Shannon entropy of the measured bitstring distribution; predictions are for noiseless preparation.

Input	$P(0000000)$	H_2 (bits)	$H_2/H_{\max}$	TV from uniform
$\lvert 0\rangle^{\otimes 7}$	0.0103	6.938	0.991	0.124
$\lvert +\rangle^{\otimes 7}$	0.0035	6.795	0.971	0.240
$\lvert K_7\rangle$	$\boxed{0.4815}$	4.274	0.611	0.584

Headline: the entropy difference between random and $|K_7\rangle$-prepared inputs is

$$\Delta H_2 \;=\; H_2(|0\rangle^{\otimes 7}) - H_2(|K_7\rangle) \;=\; 6.938 - 4.274 \;=\; \boxed{2.66 \text{ bits}}. \tag{95}$$

The entropy $H_2(|0\rangle^{\otimes 7}) = 6.938$ bits is 99.1% of the maximum $\log_2 128 = 7$ bits. All 128 bitstrings are populated, confirming that the syndrome eigenspaces are 1-dimensional and span the full Hilbert space (consistent with the K_7 stabiliser group having order $2^7 = 128$).

The $|K_7\rangle$ input gives $P(0000000) = 0.482$ vs the random-state expectation $1/128 = 0.0078$, a $62\times$ enhancement. The peak survives 112 noisy gates (56 CZ for prep, 56 CZ for uncompute); the peak's persistence directly demonstrates that $|K_7\rangle$ is preparable on hardware, distinguishable from any nearby state, and robust to depolarising noise. Back-calculated per-CZ fidelity on `ibm_fez` is 99.35%, comparable to but slightly lower than `ibm_kingston`'s 99.5%.

This verifies the syndrome-attractor reading of the Schrödinger derivation: $|K_7\rangle$ is the unique state at the all-+1 syndrome corner of the K_7 stabiliser code, and the other 127 syndrome states are accessible (they appear in the random-input distribution). Continuous syndrome measurement at the Compton-period rate would project any disturbance back to $|K_7\rangle$; this is the QEC mechanism underlying the rest-frame Schrödinger evolution $i\hbar\partial_t |K_7\rangle = m_e c^2 |K_7\rangle$ (§12).

13.10 Interpretation

The Heron R2 measurements provide hardware support for seven structural claims of the framework:

1. **The b2.13 bijection's structural content** (Paper 16 §5): the K_7 graph state $|K_7\rangle$ has $\langle H_{YY}\rangle = \dim(\mathrm{Adj}_{\mathfrak{so}(7)}) = 21$, identifying the 21 K_7 edges with the 21 $\mathfrak{so}(7)$ generators experimentally.

2. **The graph-state moment identity** (§11.3): $\langle H^n_{YY}\rangle = \dim(\mathrm{Adj})^n$ is verified at $n = 1$ on hardware (Runs 1–3). Higher moments ($n = 2$ giving 441, $n = 3$ giving 9261, etc.) are accessible through 4-qubit and 6-qubit measurement circuits, planned for follow-up runs.

3. **The bit-quantum** (§12.2): the $\dim(\mathrm{Adj}) = 21$ value of $\langle H_{YY}\rangle$ on $|K_7\rangle$ is the input to the bit-quantum derivation that gives $\kappa = m_e c^2/\dim(\mathrm{Adj})$ in the rest-frame Schrödinger equation. The hardware verification confirms this input value directly.

4. **The bracket truncation at α^2** (§11.6): Experiment 4's bulk null $\langle Y_u Y_v Y_w\rangle \approx 0$ verifies the structural 3-body identity that underlies the bracket truncation. The σ_X Furry argument is hardware-confirmed at the bulk-average level.

5. **The syndrome-attractor underpinning of the Schrödinger derivation** (§12.4): Experiment 5's $\Delta H_2 = 2.66$ bits between random and $|K_7\rangle$ inputs, with $P(0000000 \mid |K_7\rangle \text{ input}) = 0.482$ vs $1/128$ for random, confirms that $|K_7\rangle$ is the unique syndrome attractor of the K_7 stabiliser code. This is the hardware foundation for the bit-quantum derivation of $i\hbar\partial_t |K_7\rangle = m_e c^2 |K_7\rangle$.

6. **The internal-symmetry averaging principle** (§13.7): the 21 K_7 edges are statistically distinguishable on hardware ($\chi^2_{\mathrm{red}} = 30$–$61$ against edge-equality), yet the bulk $\langle H_{YY}\rangle$ identity converges within $\sim 25\%$ across all three runs. The $\mathrm{PSL}(2,7)$ symmetry of $|K_7\rangle$ is what makes this possible: the framework's predictions are sums over the internal symmetry orbit, so a hardware noise channel that locally breaks the symmetry is compatible with bulk faithfulness. This is the precise sense in which the chip realises the structure—on average over the symmetry group, not edge-by-edge.

7. **The cross-group structural identity** (§13.8): the prediction $\langle H_{YY}\rangle_{|K_N\rangle} = \binom{N}{2} = \dim(\mathrm{Adj}_{\mathfrak{so}(N)})$ extends from K_7 to K_9 on the same backend. Zero-noise extrapolation gives a K_9/K_7 ratio of 1.767 ± 0.024 vs. the predicted 1.714, agreeing to 3.07% ($+2.16\sigma$ deviation) and sharpening the cross-group bracket to $M \in [36, 39]$ at 3σ, with the structural value $M = 36$ sitting at the center. Alternatives differ by $\geq 3\sigma$; the K_7 result is not an $\mathfrak{so}(7)$ accident but a sharply falsifiable, hardware-confirmed member of the $\mathfrak{so}(2n+1)$ family.

Computation, simulation, and instantiation. A subtle point about the methodology is worth flagging. The K_7 graph state $|K_7\rangle$ appears in this paper in three guises: as a formal vector in numpy simulation (§11), as the joint $+1$-eigenstate of an abstract stabiliser algebra (§11.2), and as a configuration of Cooper pairs on a heavy-hex superconducting chip (this section). The hardware verification is therefore not external evidence about a mathematical model; it is an *instantiation* of the same mathematical object in a different physical substrate.

Furthermore, the Phase 8 derivation of the rest-frame Schrödinger equation rests on three structural inputs: Bremermann's information-processing bound, the b2.13 K_7-to-$\mathfrak{so}(7)$ bijection, and $\mathrm{PSL}(2,7)$ edge-transitive symmetry. The Heron R2 device is itself a physical system that obeys all three: Bremermann's bound constrains its information-processing rate, the K_7 graph-state preparation circuit invokes the b2.13 connectivity directly via heavy-hex routing, and the seven-qubit subspace inherits the $\mathrm{PSL}(2,7)$ symmetry of the abstract graph. In the NWT framework, the rest electron is a topological excitation of one condensate (BPS vortex defect) and a superconducting qubit is a topological excitation of another (the BCS condensate, with a Cooper-pair gap in place of the EW-scale BPS gap). Different scale, same information-theoretic framework.

The hardware experiment is therefore not just a confirmation of the graph-state moment identity; it is a demonstration that the same K_7 information structure that determines the electron's mass also determines the dynamics of a superconducting qubit register. This is the deepest implication of the experimental verification: NWT's structural unification is not theoretical in isolation but recurs at every condensate scale where topological excitations are information-processing systems.

This is, to our knowledge, the first experimental hardware support for a structural-information-theoretic claim underlying a particle-physics mass-formula derivation. The signal is clean ($+368\sigma$ above null at peak), reproducible across three backends, and inexpensive to extend (IBM Quantum Cloud academic access).

13.11 Planned follow-up experiments

The current runs verify the structural identities at $n = 0$ ($\langle 1\rangle = 1$), $n = 1$ ($\langle H_{YY}\rangle = \dim \mathrm{Adj}$), the 3-body null ($\langle Y_u Y_v Y_w\rangle = 0$ at the bulk-average level), and the syndrome-attractor uniqueness ($\Delta H_2 = 2.66$ bits). Outstanding direct tests:

- $\langle H_{YY}^2\rangle = 441$: 4-qubit measurements of $\langle Y_a Y_b Y_c Y_d\rangle$ on selected K_7 quadruples. This verifies the α^2 bracket coefficient $\dim(\mathrm{Adj})/\dim(V) = 3$ directly.
- **Stabiliser-graded subspace**: preparation of syndrome states $M < 7$ followed by single-stabiliser feedback measurements. Direct test of the syndrome-attractor reading of the rest-frame Schrödinger derivation (§12.4).
- **Additional cross-backend reproducibility**: same circuits on `ibm_fez` and other Heron R2 systems. Already verified on `ibm_kingston` (Runs 1, 2) and `ibm_marrakesh` (Run 3). Adding more devices strengthens the device-independence claim.

None of these experiments require hardware beyond current 156-qubit Heron R2 backends. Each is executable in ~ 10 minutes of quantum compute time. The complete experimental verification of the bracket structure at LO, NLO, NNLO, plus 3-body null with Fano-line resolution, is therefore within reach in the near term.

14 The closed formula and CODATA precision

Sections 2–9 derived each coefficient in the boxed expression (7) from Spin(7) representation theory and octonion algebra, and §11 re-derived the bracket coefficients information-theoretically as the first two moments of H_{YY} on $|K_7\rangle$. This section assembles the pieces, evaluates the formula at the CODATA value of α, and tests the resulting predictions against $m_e/m_{\rm Pl}$ and G. Read through the QEC spine, the closed formula (102) is the *fidelity expansion* of the encoded electron worldtube on $|K_7\rangle$ to second order in interaction-event noise α.

14.1 Assembly of the closed formula

Collecting the identifications from preceding sections:

$$\text{classical prefactor} \quad \frac{\dim S}{\dim V} = \frac{8}{7} \qquad (\S6) \tag{96}$$

$$\text{linear-in-}\alpha\text{ bracket term} \quad \frac{1}{\dim V} = \frac{1}{7} \qquad (\S7) \tag{97}$$

$$\alpha^2\text{ bracket term} \quad \frac{\dim \mathrm{Adj}}{\dim S} = \frac{21}{8} \qquad (\S8) \tag{98}$$

$$\text{per-edge RT amplitude} \quad \sin\left(\frac{\pi}{k+h^\vee}\right) = \sqrt{\alpha} \qquad (\S3) \tag{99}$$

$$\text{number of } K_7 \text{ edges} \quad |E(K_7)| = 21 \qquad (\S3) \tag{100}$$

$$\text{Paper 8a dictionary} \quad \sin\left(\frac{\pi}{k+h^\vee}\right) = \frac{1}{2^{1/4}\,\kappa} \qquad (\S4). \tag{101}$$

Substituting these back into the RT Wilson amplitude on the Heegaard-torus embedding of K_7 gives the closed form

$$\frac{m_e}{m_{\rm Pl}} = \frac{\dim S}{\dim V}\left[1 + \frac{\alpha}{\dim V} + \frac{\dim \mathrm{Adj}}{\dim S}\,\alpha^2\right]\sin^{21}\left(\frac{\pi}{k+h^\vee}\right). \tag{102}$$

Equivalently, since $\sin(\pi/(k+h^\vee)) = \sqrt{\alpha}$ under the Paper 8a dictionary, the bracket expansion gives

$$\frac{m_e}{m_{\rm Pl}} = \frac{8}{7}\left(1 + \frac{\alpha}{7} + \frac{21}{8}\,\alpha^2\right)\alpha^{21/2}. \tag{103}$$

At the CODATA value $\alpha = 7.297\,352\,5693\times 10^{-3}$, the Chern–Simons level determined by the dictionary is $k = \pi/\arcsin(\sqrt{\alpha}) - h^\vee = 31.7314$.

14.2 Precision test: $m_e/m_{\rm Pl}$

Table 5 compares the prediction of (103) at each order in α against the CODATA value $m_e/m_{\rm Pl} = 4.185\,44 \times 10^{-23}$ [13].

Table 5: $m_e/m_{\rm Pl}$ at each order of the closed formula vs CODATA. The NNLO prediction improves agreement by $\sim$140$\times$ over Paper 14's NLO.

Order	Expression	Deviation vs CODATA	
LO	$\frac{8}{7}\,\alpha^{21/2}$	-0.1181%	
NLO	$\frac{8}{7}\left(1+\frac{\alpha}{7}\right)\alpha^{21/2}$	-0.0140%	(Paper 14)
NNLO	$\frac{8}{7}\left(1+\frac{\alpha}{7}+\frac{21}{8}\,\alpha^2\right)\alpha^{21/2}$	-0.0001%	(this work)

The NNLO agreement is at the 1-part-in-10^6 level—below the precision of the CODATA value of $m_e/m_{\rm Pl}$ itself, which is inherited from the experimental determination of G (see §14.4).

14.3 Induced prediction for Newton's constant

The Planck mass is defined by $m_{\mathrm{Pl}} = \sqrt{\hbar c/G}$, so (103) implies

$$G = \frac{\hbar c}{m_e^2}\left[\frac{8}{7}\right]^2\left(1+\frac{\alpha}{7}+\frac{21}{8}\,\alpha^2\right)^2\alpha^{21}. \tag{104}$$

The leading term $(8/7)^2\,\alpha^{21}\,\hbar c/m_e^2$ reproduces the leading order of Paper 16's identity. At CODATA values of $\hbar$, c, m_e, and α, table 6 compares the order-by-order prediction against $G = 6.674\,30(15)\times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ [13].

Table 6: G at each order of the closed formula vs CODATA ($G_{\mathrm{exp}} = 6.674\,30\times10^{-11}$, 22 ppm precision). The NNLO prediction lands within the experimental uncertainty on G itself.

Order	Expression	G_{pred}	Dev. vs CODATA	
LO	$\left(\frac{8}{7}\right)^2\alpha^{21}$	6.6585×10^{-11}	-0.237%	
NLO	$\left(\frac{8}{7}\right)^2\left(1+\frac{\alpha}{7}\right)^2\alpha^{21}$	6.6724×10^{-11}	-0.029%	
NNLO	$\left(\frac{8}{7}\right)^2\left(1+\frac{\alpha}{7}+\frac{21}{8}\alpha^2\right)^2\alpha^{21}$	6.6742×10^{-11}	-0.0011%	(11 ppm)

The NNLO prediction is 11 ppm below the central CODATA value, well inside the 22 ppm experimental uncertainty. In this regime the closed formula (104) is constrained *by* the experimental determination of G rather than constraining it.

14.4 The precision limit is set by G

Every input to the formula except G is known to higher precision than G itself:

- c and $\hbar$ are exact post-2019 SI definitions.
- m_e is known to $\sim 3\times10^{-11}$ relative precision [13].
- The fine structure constant α is known from the electron g-factor to $\sim 8\times10^{-11}$ precision [13].

The bottleneck is G, which shows inter-experiment dispersion of order 100–300 ppm among modern torsion-balance measurements [14, 15] and carries 22 ppm in the CODATA 2018 recommended value. Propagating through $m_{\mathrm{Pl}} = \sqrt{\hbar c/G}$, the uncertainty on m_e/m_{Pl} is ~ 11 ppm, which in turn limits what we can infer about the α^2 coefficient c_2 in (103). The α^2 contribution to m_e/m_{Pl} at CODATA α is of order 1.6×10^{-4}, so a c_2-shift of order unity changes m_e/m_{Pl} by $\sim 5\times10^{-5}$. CODATA m_e/m_{Pl} therefore determines c_2 to ±0.2 ($\sim$7%). The structural identification $c_2 = \dim(\mathrm{Adj})/\dim(V) = 3$ established in §8 matches the empirical value $c_2 = 3.01$ to $\sim$0.5%, far inside the CODATA-limited window of $\pm$7%.

In other words, the 0.5% residual between empirical and structural c_2 is consistent with the CODATA precision limit on m_e/m_{Pl}; a future determination of G at, say, 1–10 ppm precision (via atom interferometry or LISA-scale gravitational wave metrology) would sharpen this test, with any deviation beyond the CODATA bar constituting a falsification of the $c_2 = \dim(\mathrm{Adj})/\dim(V)$ identification.

15 The structural minimum: one absolute scale + topology

The framework presented in this paper has a striking parsimony property: within NWT, the remaining physical content reduces to topological data plus a single absolute mass scale, with all unit-conversion constants taken as conventions. This section makes the parsimony claim precise. The strongest defensible reading is that the model achieves a strong form of internal parsimony — one absolute mass scale plus topological and Lie-theoretic data — once the framework's normalisation choices are fixed. We do not claim that no further reduction is possible in any conceivable theory, only that this is the strongest parsimony statement presently supported within NWT.

15.1 Why exactly three dimensional constants are required

Physics is described in four-dimensional spacetime with three fundamental dimensions: mass M, length L, and time T. Setting units in these three dimensions requires *exactly three independent dimensional constants*, no more and no fewer. This is not a feature of a particular theory; it is forced by the dimensional structure of physical observables.

A useful choice is the set $\{\hbar, c, v_{EW}\}$:

$$[\hbar] = ML^2/T, \qquad [c] = L/T, \qquad [v_{EW}] = ML^2/T^2 \text{ (energy)}. \tag{105}$$

These are linearly independent in the dimensional vector space spanned by $\{M, L, T\}$, and together they fix unit conversion in all three dimensions. Any other dimensional quantity of physical significance — the Planck length $\ell_{\rm Pl}$, the electron mass m_e, the gravitational coupling G, etc. — is a derived combination of these three.

A natural temptation is to reduce the input set to two: for instance, can the speed of light c be derived from $\{v_{EW}, \ell_{\rm Pl}\}$ alone? The answer, by direct dimensional analysis, is **no**. With $\{v_{EW}, \ell_{\rm Pl}, \hbar\}$ as inputs:

$$T = \hbar/v_{EW}, \tag{106}$$
$$L = \ell_{\rm Pl}, \tag{107}$$
$$c = L/T = \ell_{\rm Pl}\, v_{EW}/\hbar. \tag{108}$$

Numerically:

$$\ell_{\rm Pl}\, v_{EW}/\hbar = \frac{(1.616\times 10^{-35}\ \mathrm{m})(3.94\times 10^{-8}\ \mathrm{J})}{1.055\times 10^{-34}\ \mathrm{J{\cdot}s}} \approx 6.07\times 10^{-9}\ \mathrm{m/s}, \tag{109}$$

which is 5×10^{16} times smaller than the actual $c \approx 3\times 10^8$ m/s. The "missing factor" equals $m_{Pl}c^2/v_{EW} \approx 5\times 10^{16}$, the inverse of NWT's electroweak-Planck hierarchy prediction $v_{EW}/m_{Pl}c^2 \approx 2\times 10^{-17}$ (§11.8). Restoring this factor yields

$$c \;=\; \ell_{\rm Pl}\cdot v_{EW}/\hbar \times m_{Pl}c^2/v_{EW} \;=\; \ell_{\rm Pl}\cdot m_{Pl}c^2/\hbar \;=\; \ell_{\rm Pl}\cdot \hbar c/(\ell_{\rm Pl}\hbar) \;=\; c, \tag{110}$$

which is tautological. The reason is that $\ell_{\rm Pl} = \sqrt{\hbar G/c^3}$ already contains c implicitly. Every "fundamental length" in physics — Planck length, Compton wavelength, BPS healing length — is built from masses, $\hbar$, and c. There is no length scale in physics independent of c, so c cannot be eliminated by any choice of length input.

Three is the structural minimum.

15.2 NWT achieves the minimum: one physical input + conventions

If three dimensional constants are required, the parsimony question is: how many of them are *physical inputs*, and how many are *unit conventions*?

In post-2019 SI, four constants are exact-by-definition:

$$\hbar, \quad c, \quad k_B, \quad e \quad \text{(post-2019 SI: exact)}. \tag{111}$$

These define the kilogram, metre, kelvin, and ampere respectively. They are not measured quantities; their numerical values are chosen to maintain continuity with previous SI definitions.

Within NWT, every dimensionless ratio in observed physics is derived from topological data alone:

- $\alpha = 1/(25\pi\sqrt{2}\kappa^2)$ from the trefoil's Aharonov–Bohm phase (Paper 8a);
- All particle mass ratios m_X/m_e from torus knot mode-locking (Paper 6, 1.06% median);
- All gauge couplings and mixing angles from BPS topology (Papers 8a, 13);
- m_e/m_{Pl} from the K_7 Wilson amplitude (§14, 0.0001%);
- v_{EW}/m_{Pl} from the combination of Paper 6 (y_e) and this paper's m_e/m_{Pl};
- Rest-frame Schrödinger evolution $i\hbar\partial_t |K_7\rangle = m_e c^2 |K_7\rangle$ from BPS + b2.13 + Bremermann + PSL$(2,7)$ (§12).

Combining these ingredients with the four SI exact constants leaves exactly *one* physical input free: the absolute mass scale. The conventional choice is $v_{EW} = 246.22$ GeV (measured at the LHC), but $m_e = 0.511$ MeV or any other particle mass would do equally well, since all are related by NWT-predicted dimensionless ratios.

$$\boxed{\begin{aligned} \text{NWT input set} \;=\; \{\,\text{topological data}\,\} \;+\; \{\,\text{one absolute mass scale}\,\} \\ +\; \{\,\hbar,\, c,\, k_B,\, e\ \text{(SI exact)}\,\} \end{aligned}} \tag{112}$$

This is, within NWT, a strong form of internal parsimony. Three dimensional constants are required by dimensional analysis; NWT places three of them ($\hbar$, c, k_B; equivalently e and ratios) as unit conventions, and the remaining one (v_{EW}) is the only absolute physical input.

15.3 Comparison to the Standard Model

The Standard Model has approximately 30 free parameters: nine fermion masses (or equivalently nine Yukawa couplings), three gauge couplings, four CKM parameters, four PMNS parameters, the Higgs mass and self-coupling, the QCD theta angle, and the electroweak vacuum expectation value v_{EW}. All of these are treated as inputs: measured experimentally without theoretical derivation.

NWT proposes a topologically-derived dimensionless expression for each of these parameters (Papers 6, 8a, 13, this paper). Within the framework, the single remaining absolute input is the scale v_{EW}, which sets the conversion to SI units. The Standard Model's ~ 30 free parameters can be organised around one absolute scale plus topological and Lie-theoretic data:

$$\text{SM:} \;\; \sim 30 \text{ free parameters} \quad \longrightarrow \quad \text{NWT:} \;\; 1 \text{ absolute scale} \;+\; \text{topology}. \tag{113}$$

The reduction by a factor of ~ 30 in parameter count is the quantitative measure of NWT's parsimony. The reduction is not achieved by hiding parameters in undetermined coefficients — as demonstrated in §§2–13, every NWT prediction is a specific topological calculation with no free parameters at the dimensionless level.

15.4 What further reduction would require

The structural minimum is one absolute scale. Any framework attempting to eliminate this last input must derive the SI numerical value of v_{EW} (or equivalently, m_e, m_p, etc.) from purely topological data, without reference to any externally-fixed unit. This is dimensionally impossible: the SI unit system requires *some* definition of mass (or energy or length or time), and the kilogram is currently defined via $\hbar$ and the cesium-133 hyperfine transition frequency. Any "derivation" of v_{EW} in SI units necessarily reduces to "v_{EW} measured against some chosen unit standard."

A more interesting reduction would be a *Volovik-style* emergent-c derivation that connects c to the underlying condensate structure, predicting c_s as a Bogoliubov phonon speed in a non-relativistic underlying theory whose long-wavelength limit is NWT's relativistic Lagrangian. This would not eliminate c as a unit (which is impossible), but would predict *Lorentz-violation signatures* at the BPS scale of order $\Delta c/c \sim (m_e/m_H)^2/4 \approx 1.7 \times 10^{-11}$. Below current experimental bounds but testable with future precision improvements.

The Volovik refinement would not change the parameter count; it would add a falsifiable prediction. The qualitative breakthrough of NWT — one absolute scale plus topology — is established by the framework presented in this paper, with no Volovik-style extension required.

15.5 Hardware support for the structural minimum

The IBM Heron R2 measurements reported in §13 provide hardware support for the topological data — the K_7 graph-state structure and the b2.13 bijection — on real quantum hardware. Three runs across two backends measure $\langle H_{YY}\rangle$ values consistent with $\dim(\mathrm{Adj}_{\mathfrak{so}(7)}) = 21$ within hardware-noise expectations, with $+368\sigma$ separation from any random-state null hypothesis.

This is significant for the parsimony claim: if the *topological data* of NWT can be probed directly on a quantum computer, then the framework rests on a foundation that is experimentally accessible rather than only inferable from particle-physics phenomenology. The Standard Model's ~ 30 free parameters can be organised around one absolute scale plus a topological-data structure that is itself experimentally accessible on near-term quantum hardware.

16 Discussion and outlook

16.1 What is supported experimentally

The measurement of $\langle H_{YY}\rangle = 16.4715 \pm 0.0448$ on IBM Heron R2 (§13, Run 2) and the Run 1 value 16.00 ± 0.21 provide hardware support for the structural identity $\langle H_{YY}\rangle_{|K_7\rangle} = \dim(\mathrm{Adj}_{\mathfrak{so}(7)}) = 21$ within hardware-noise expectations, with $+368\sigma$ separation from any random-state null hypothesis. Because this identity is the input to (a) the $\alpha/7$ NLO bracket coefficient in m_e/m_{Pl}, (b) the bit-quantum derivation of rest-frame Schrödinger evolution, and (c) the b2.13 K_7-to-$\mathfrak{so}(7)$ bijection underlying Paper 16, the Heron measurement provides direct experimental support for these three pillars of the QEC framework simultaneously.

The signal is $+76\sigma$ above the random-state null hypothesis and the back-calculated per-CZ fidelity (99.47%) matches the published Heron R2 specification, leaving no ambiguity about the cause of the 24% reduction from the noiseless prediction.

This is, to our knowledge, the first experimental verification of a structural identity from a particle-physics mass-formula derivation on programmable quantum hardware. Follow-up runs on the same hardware class can verify the higher moments ($\langle H_{YY}^2\rangle = 441$ via 4-qubit measurements, $\langle Y_aY_bY_c\rangle = 0$ via 3-qubit measurements) directly, testing the bracket truncation at α^2 as an experimental prediction.

16.2 What is derived within the framework

The structural framework presented here organises the chain from the K_7 Wilson-graph carrier to the CODATA value of $m_e/m_{\rm Pl}$, under the framework's normalisation choices and with α taken as input from Paper 8a's Aharonov–Bohm calculation. At LO, NLO, and NNLO, the resulting closed form matches CODATA $m_e/m_{\rm Pl}$ at the $10^{-4}\%$ level. Specifically:

- The information conservation theorem (§11.1) is derived from energy conservation, zero superfluid viscosity, and the Bekenstein– Shannon information content of configurational degrees of freedom. It forces KE to transduce into discrete topological information for stable solitons.
- The $\sqrt{\alpha}$-per-edge rule (§3.3) is derived from the Spin(7) RT R-matrix in the adjoint channel of $V \otimes V$, with Drinfeld exponent $C_{\rm Adj} - 2C_V = -2$.
- The $k \leftrightarrow \kappa$ dictionary (§4) is fixed by the AB-phase calculation on the BPS vortex tube of Paper 8a [31].
- The exponent 21/2 in $\alpha^{21/2}$ is RT-natural for Spin(7) at any level k (§3.8): the modular total dimension scales as $\mathcal{D}^2 \sim (k + h^\vee)^{\dim \mathfrak{g}}$, and the per-edge factor as $(k + h^\vee)^{-1}$, so the product $\mathcal{D}^2 \cdot \alpha^{|E(K_7)|/2}$ is level-independent precisely when $|E(K_7)| = \dim \mathfrak{g}$, which holds for $\mathfrak{so}(7)$ by the Paper 16 b2.13 bijection.
- The choice of vector representation V for the K_7 Wilson line is structurally forced (§5): no irreducible component of Adj $\otimes$ Adj has the requisite Drinfeld exponent -2, so the alternative line representation fails to reproduce the per-edge $\sqrt{\alpha}$.
- The classical prefactor $\dim(S)/\dim(V) = 8/7$ (§6) is fixed by the Spin(7) spinor-vector branching at the worldline endpoints (Paper 16 b2.14).
- The bracket coefficients $\alpha/\dim(V)$ and $\dim(\mathrm{Adj})/\dim(V)\cdot\alpha^2$ are exactly the first two normalised moments of H_{YY} on the K_7 stabiliser graph state $|K_7\rangle$ (§11.4), with the structural identity $\langle H_{YY}^n \rangle_{|K_N\rangle} = \dim(\mathrm{Adj}_{\mathfrak{so}(N)})^n$ verified across the $\mathfrak{so}(2n+1)$ family at $N \in \{7, 9, 11\}$.
- The bracket truncation at α^2 is supported (§11.6) by the 3-body operator probe $\langle Y_a Y_b Y_c \rangle_{|K_7\rangle} = 0$ for all 35 basis triples, consistent with a representation-theoretic suppression mechanism rather than accidental cancellation; a fully rigorous derivation of the mechanism remains open.
- The octonion identity $\sum |[\cdot]|^2 = 112$ (§9) is derived from the composition law $|[a, b, c]|^2 = 4(\det g - |\varphi|^2)$ of the seven imaginary octonions; this identity, combined with the Casimir balance of the Adj/V ratio, controls the α^2 coefficient.

16.3 What is identified but not yet proved

The bracket $[1 + \alpha/\dim(V) + (\dim \mathrm{Adj}/\dim V)\,\alpha^2]$ is *structurally identified* via two complementary routes: the chain of representation-theoretic considerations of §7, §8, and §9; and the graph-state moment identity $\langle H_{YY}^n \rangle_{|K_7\rangle} = \dim(\mathrm{Adj}_{\mathfrak{so}(7)})^n$ of §11. Each route reduces every coefficient to a Spin(7) integer with no fitted parameter, and the closed-form match to CODATA $m_e/m_{\rm Pl}$ at -0.0001% corroborates the identification at the precision of the input α.

What is *not* provided is a single direct RT manipulation in $\mathrm{Spin}(7)_{k=32}$ that produces the full bracket from a chain of $6j$-symbol or modular identities. Three independent attempts were undertaken in the course of this work—Schur–Racah reduction of the K_7 spin network, Heegaard modular-S ansätze, and a Lawrence–Rozansky surgery sum on the trefoil ($\Sigma(2, 3, 5)$ realised as -1 surgery on

$T(2,3)$). None reproduces the bracket at small α at our level of computation: the S^2 Verlinde V_n amplitude is α-independent (a topological count of $V^{\otimes n} \to \mathbf{1}$ invariants), and the Heegaard / surgery sums have $|Z|/\alpha^{21/2}$ that diverges with k rather than approaching a finite constant carrying the bracket's coefficients.

We interpret the failure of these three routes as substantive: it suggests the rigorous RT derivation requires either (a) the full Reshetikhin–Kirillov asymptotic $6j$ machinery for B_3 with explicit $\mathrm{Sym}^2/\wedge^2$ Frobenius–Schur indicators, or (b) the rank-3 Lawrence–Rozansky perturbative expansion for $\Sigma(2,3,5)$ in $\mathrm{Spin}(7)_k$, neither of which is available in standard sources and both of which require non-trivial development. The graph-state derivation of §11 provides a third route that *does* reproduce the bracket coefficients exactly at LO and NLO, but does so via the moment structure of $|K_7\rangle$ as a stabiliser state, not via an RT-internal derivation; the connection between the graph-state expectation and the underlying RT amplitude remains to be made explicit.

16.4 Open problems

The truncation mechanism at α^2. The 3-body operator probe of §11.6 shows that $\langle Y_a Y_b Y_c \rangle_{|K_7\rangle} = 0$ exactly for all 35 basis triples, supporting truncation of the bracket at α^2 through a representation-theoretic mechanism rather than accidental cancellation. We further show (§11.6) that the 3-body vanishing is enforced by a Furry-like global X-parity $\sigma_X = \prod_v X_v$ acting on $|K_7\rangle$. This is consistent with a representation-theoretic mechanism for the truncation, but the same Furry parity *commutes* with H_{YY} (each two-body $Y_u Y_v$ pair is σ_X-even), so it does not by itself constrain $\langle H_{YY}^n \rangle$ for any n; the precise mechanism that sources the truncation at α^2 in the full RT calculation remains open.

Direct numerical computation (`nwt_truncation_mechanism.py`) further rules out the simplest version of the Casimir-hierarchy candidate: the classical-limit Casimir eigenvalues on V are $C_2(V) = 6$, $C_4(V) = 111$, $C_6(V) = 1594.125$, all non-vanishing, so $C_6(V) = 0$ at the classical level cannot be the truncation source. This narrows the candidate space to:

- (a′) A level-k refinement of the Casimir hierarchy: at level $k = 32$, fusion-rule integrability constraints may cancel the α^3 Wilson-amplitude contribution despite $C_6(V)$ being non-zero in the classical limit.
- (c) Spinor–vector branching $S = V \oplus 1$ at the Wilson endpoints (§6, Paper 16 b2.14): rank-1 mismatch between $\dim(S) = 8$ and $\dim(V) = 7$ may allow only quadratic mixing in α.

The cleanest test of (c) is to compute the perturbative expansion of the boundary projector $\Pi_{S\to V}$ at order α^n and verify truncation at α^2. The cleanest test of (a′) is a targeted α^3 coefficient calculation in the Lawrence–Rozansky perturbative expansion for $\Sigma(2,3,5)$ at $\mathrm{Spin}(7)_k$ (asking whether $a_3 = 0$ or $a_3 = 63$, the geometric prediction). Either test would resolve the truncation mechanism; both require non-trivial RT machinery beyond the scope of this paper.

A partial probe of option (c) at the q-deformed-dimension level (`nwt_truncation_qdef.py`) finds that the simplest reading of the G_2 branching is broken at level k. Numerical fits at

$$k \in \{10,\ 16,\ 24,\ 32,\ 50,\ 75,\ 100,\ 200,\ 500\}$$

give

$$\dim_q(S) - \dim_q(V) - 1 \ \approx\ -11.81\,\alpha^2 + 9.35\,\alpha^3 + 1.51\,\alpha^4 + \cdots, \tag{114}$$

with non-vanishing α^3 and α^4 coefficients. The classical relation $\dim(S) = \dim(V) + 1 = 8$ does therefore *not* lift exactly to the q-deformation as a clean polynomial of degree 2, although the α^2 coefficient dominates at small α. This rules out the simplest dimension-level reading of option (c); a refined version, in which the boundary projector $\Pi_{S\to V}$ has structure beyond just the dimension ratio (e.g., quantum Clebsch–Gordan coefficients with their own truncation properties), is still tenable but requires deeper computation.

A partial probe of option (a′) finds that the level-k fusion-channel structure of $V \otimes V$ has all three channels (singlet, Adj, and **27**) contributing at finite k, with amplitudes $|1 + R_{\mathbf{1}}|/2 \approx 0.87$, $|1 + R_{\mathrm{Adj}}|/2 \approx 0.085 \approx \sqrt{\alpha}$, and $|1 + R_{\mathbf{27}}|/2 \approx 1.0$ at $k = 32$. The Paper 17 reduction to the Adj channel at each edge gives $\alpha^{21/2}$, but the full amplitude includes contributions from the other channels that may generate the bracket structure. An α^3 cancellation between these channel contributions would manifest option (a′); numerical verification requires the full Lawrence–Rozansky calculation.

Neither partial probe is decisive. The truncation mechanism remains the central representation-theoretic open problem of this paper.

16.5 Bracket factorisation and a heuristic for truncation

The bracket admits a clean structural factorisation (`nwt_truncation_channel_mixing.py`):

$$\mathrm{bracket}(\alpha) \;=\; 1 + \frac{\alpha}{\dim V} + \frac{\dim(\mathrm{Adj})}{\dim V}\,\alpha^2 \;=\; 1 + \frac{\alpha}{\dim V}\big(1 + \dim(\mathrm{Adj})\,\alpha\big). \tag{115}$$

For $\mathfrak{so}(7)$: $1 + (\alpha/7)(1 + 21\alpha)$. The factor $(1 + 21\alpha)$ is exactly the first-order Taylor expansion of $1/(1 - \dim(\mathrm{Adj})\,\alpha) = 1/(1 - 21\alpha)$. In other words, the bracket equals the first two terms of the geometric series $1 + \alpha/\big[\dim V\,(1 - \dim(\mathrm{Adj})\,\alpha)\big]$:

$$\frac{1}{\dim V\,(1 - \dim(\mathrm{Adj})\,\alpha)} \;=\; \frac{1}{\dim V}\big[1 + \dim(\mathrm{Adj})\,\alpha + (\dim(\mathrm{Adj})\,\alpha)^2 + \cdots\big], \tag{116}$$

truncated at the term linear in $\dim(\mathrm{Adj})\,\alpha$.

This factorisation suggests a heuristic for the truncation mechanism. In the QFT-style perturbative reading of the K_7 Wilson amplitude, the $\alpha/\dim(V)$ factor is a propagator-like term (one V exchange weighted by $1/\dim V$), and the $(1 + \dim(\mathrm{Adj})\,\alpha)$ factor counts insertions of $\mathfrak{so}(7)$ generators on the V-line:

- α^0: no generator insertion (tree-level).
- α^1: one generator inserted, with $\dim(\mathrm{Adj}) = 21$ choices.
- $\alpha^{n\geq 2}$: multiple generator insertions on a single edge — *forbidden by the b2.13 bijection* (each K_7 edge carries exactly one $\mathfrak{so}(7)$ generator).

Under this reading, the bracket truncates at α^2 because b2.13's edge-labeling rule enforces a single-generator-per-edge constraint, which is mathematically the same as truncating the geometric series at first order in $(\dim(\mathrm{Adj})\,\alpha)$. The geometric series prediction for α^3 and higher would correspond to multi-generator insertions on a single edge, which are inconsistent with the b2.13 bijection.

This heuristic is not yet a derivation: a rigorous proof would require deriving the b2.13 single-generator-per-edge rule from the underlying Reshetikhin–Turaev formalism (presumably from the V–Adj fusion structure of $V \otimes V$ at level k, with the adjoint-channel projection only allowing one generator at each edge). But the factorisation (115) identifies the precise structural form of the truncation: the bracket is the first two terms of a finite geometric expansion, with higher terms cancelled by the b2.13 single-generator constraint. This is the most concrete characterisation of the truncation mechanism currently available, and a clear target for future RT-formal derivation.

The bracket-derivation target. The first-principles RT derivation of the bracket $[1+\alpha/\dim V + (\dim \mathrm{Adj}/\dim V)\,\alpha^2]$ from $\mathrm{Spin}(7)_{k=32}$ data is the central RT-formal open problem. Section 11 provides a graph-state derivation that reproduces the bracket coefficients exactly, but the connection between the graph-state moment identity and the underlying RT amplitude is not yet rigorous. A successful derivation would:

1. Replace the structural identifications of §§7–9 with a single RT-formal manipulation (or, equivalently, derive $\langle H_{YY}^n\rangle = \dim(\mathrm{Adj})^n$ from RT data).
2. Provide an RT analog of the 3-body truncation result of §11.6, identifying which of the representation-theoretic mechanisms (Casimir, Furry, or branching) is responsible.
3. Connect the per-vertex factor $\alpha/\dim(V)^2$ and the α^2 coefficient $\dim(\mathrm{Adj})/\dim(V)$ to a unified RT identity, rather than independent structural matches.

We expect such a derivation to proceed via the asymptotic expansion

$$\mathcal{Z}_{\mathrm{Spin}(7)_k}(\Sigma(2,3,5), K_7 \text{ in } V) \;=\; (8/7)\cdot\alpha^{21/2}\cdot\left[1+a_1\alpha+a_2\alpha^2\right] \tag{117}$$

in the perturbative-CS framework of Witten–Reshetikhin–Lawrence, with the bracket coefficients a_n emerging as Casimir-rational combinations of $\mathfrak{so}(7)$ structural invariants matching the graph-state moments. This is the natural target of a successor paper.

Higher-order α corrections. The 3-body probe shows that the bracket truncates at α^2, so any α^3 residual in the data must come from $\alpha^{21/2}$-power corrections (e.g., from α^3 in $\alpha^{21/2}$ rather than α^3 in the bracket) or from a higher-loop gauge correction not captured by the K_7 graph state. Sharpening the CODATA value of G (currently 22 ppm) would expose any such residual and constrain the source.

The fine-structure constant. This paper takes α as input from Paper 8a's vortex-tube Aharonov–Bohm calculation $1/\alpha = 25\pi\sqrt{3}+1$, which is itself a structural identification rather than a rigorous field-theoretic derivation. A first-principles derivation of α from $\mathfrak{so}(7)$ coupling structure would close the last input parameter of the gravitational chain.

Generalisation beyond Spin(7). The modular-naturalness condition $|E(K_N)| = \dim\mathfrak{g}$ holds for any $\mathfrak{so}(N)$ with the corresponding K_N graph (§3.8). The graph-state moment identity also extends across the $\mathfrak{so}(2n+1)$ family (§11.5, verified at K_9 and K_{11}). Whether a higher-rank topological input (e.g., a torus knot $T(p,q)$ with $p,q \neq 2,3$) would give an analogous closed-form expression for a hypothetical higher-rank gauge group is an open question. In the rank-3 case ($\mathfrak{so}(7)$, this paper), the Cartan-graded subspace prefactor reading $\dim(S)/(\dim(S)-1)$ coincides with the spinor–vector boundary projection $\dim(S)/\dim(V)$ via the rank-3 degeneracy $\dim(V)=\dim(S)-1$; at higher rank these readings would differ, providing a falsifiable cross-check on which prefactor identification is fundamental.

Information-theoretic synthesis paper. The information-theoretic spine of this paper invites a fuller treatment in a successor work: deriving the conservation theorem (57) from a rigorous version of Verlinde–Padmanabhan emergent gravity adapted to discrete topological sectors, identifying the natural quantum-information metric on the configurational degrees of freedom of the NWT condensate, and connecting the K_7 graph-state encoding to the physical worldtube of the electron in $S^3/2I$ (readings (A) and (B) of §11.9). The numerical results of this paper provide a quantitative anchor for that program.

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Code availability

All analysis code and reproducibility data for the calculations and hardware experiments reported in this paper are available at https://github.com/JimGalasyn/null-worldtube, including:

- K_N graph-state moment numerics:
  `nwt_qec_bracket_test.py`,
  `nwt_qec_KN_generalization.py`
- Bracket-truncation probes:
  `nwt_truncation_mechanism.py`,
  `nwt_truncation_qdef.py`,
  `nwt_truncation_channel_mixing.py`
- IBM Heron R2 submission and analysis:
  `nwt_qec_heron_*.py`
- $\mathrm{PSL}(2,7)$ edge-transitivity re-analysis:
  `nwt_qec_psl27_edge_transitivity.py`
- Forward-prediction and zero-noise extrapolation:
  `nwt_qec_forward_prediction.py`,
  `nwt_qec_zne_*.py`,
  `nwt_qec_heron_zne.py`
- Raw Heron job outputs:
  `analysis/heron_results/2026-04-26_*.txt`

References

[1] E. Witten. *Quantum field theory and the Jones polynomial.* Commun. Math. Phys. **121**, 351 (1989).

[2] N. Reshetikhin and V. G. Turaev. *Invariants of 3-manifolds via link polynomials and quantum groups.* Invent. Math. **103**, 547 (1991).

[3] V. G. Kac and D. H. Peterson. *Infinite-dimensional Lie algebras, theta functions, and modular forms.* Adv. Math. **53**, 125 (1984).

[4] D. S. Freed and R. E. Gompf. *Computer calculation of Witten's 3-manifold invariant.* Commun. Math. Phys. **141**, 79 (1991).

[5] L. C. Jeffrey. *Chern–Simons–Witten invariants of lens spaces and torus bundles, and the semi-classical approximation.* Commun. Math. Phys. **147**, 563 (1992).

[6] R. Lawrence and L. Rozansky. *Witten–Reshetikhin–Turaev invariants of Seifert manifolds.* Commun. Math. Phys. **205**, 287 (1999).

[7] J. M. F. Labastida and C. Lozano. *Lectures on topological quantum field theory.* AIP Conf. Proc. **419**, 54 (1998).

[8] L. Euler. *Solutio problematis ad geometriam situs pertinentis.* Commentarii Academiae Scientiarum Imperialis Petropolitanae **8**, 128 (1736).

[9] C. Hierholzer and C. Wiener. *Über die Möglichkeit, einen Linienzug ohne Wiederholung und ohne Unterbrechung zu umfahren.* Math. Ann. **6**, 30 (1873).

[10] P. J. Heawood. *Map-colour theorem.* Quart. J. Pure Appl. Math. **24**, 332 (1890).

[11] G. Ringel and J. W. T. Youngs. *Solution of the Heawood map-colouring problem.* Proc. Natl. Acad. Sci. USA **60**, 438 (1968).

[12] J. C. Baez. *The Octonions.* Bull. Amer. Math. Soc. **39**, 145 (2002).

[13] E. Tiesinga, P. J. Mohr, D. B. Newell, and B. N. Taylor. *CODATA recommended values of the fundamental physical constants: 2018.* Rev. Mod. Phys. **93**, 025010 (2021).

[14] G. Rosi, F. Sorrentino, L. Cacciapuoti, M. Prevedelli, and G. M. Tino. *Precision measurement of the Newtonian gravitational constant using cold atoms.* Nature **510**, 518 (2014).

[15] L.-C. Tu, Q. Li, Q.-L. Wang, C.-G. Shao, S.-Q. Yang, L.-X. Liu, Q. Liu, and J. Luo. *New determination of the gravitational constant G with time-of-swing method.* Phys. Rev. D **82**, 022001 (2010).

[16] E. Verlinde. *On the Origin of Gravity and the Laws of Newton.* J. High Energy Phys. **2011**, 029 (2011), arXiv:1001.0785.

[17] T. Padmanabhan. *Thermodynamical Aspects of Gravity: New Insights.* Rep. Prog. Phys. **73**, 046901 (2010), arXiv:0911.5004.

[18] H. J. Bremermann. *Quantum Noise and Information.* In: *Proc. Fifth Berkeley Symp. on Mathematical Statistics and Probability*, vol. 4, pp. 15–20, Univ. California Press (1967).

[19] J. D. Bekenstein. *Black Holes and Entropy.* Phys. Rev. D **7**, 2333 (1973).

[20] C. E. Shannon. *A Mathematical Theory of Communication.* Bell Syst. Tech. J. **27**, 379–423 and 623–656 (1948).

[21] W. H. Furry. *A Symmetry Theorem in the Positron Theory.* Phys. Rev. **51**, 125 (1937).

[22] V. G. Drinfeld. *Quantum Groups.* In: *Proc. Int. Congress of Mathematicians* (Berkeley, 1986), vol. 1, pp. 798–820, Amer. Math. Soc. (1987).

[23] E. Brieskorn. *Beispiele zur Differentialtopologie von Singularitäten.* Invent. Math. **2**, 1–14 (1966).

[24] H. S. M. Coxeter. *Regular Polytopes.* 3rd ed., Dover (1973).

[25] H. Freudenthal. *Zur Berechnung der Charaktere der halbeinfachen Lieschen Gruppen, I.* Indag. Math. **16**, 369–376 (1954).

[26] R. Brauer. *On Algebras Which are Connected with the Semisimple Continuous Groups.* Ann. Math. **38**, 857–872 (1937).

[27] J. P. Galasyn and C. Théodore. *Self-Consistent Electron from Phase Closure on a Torus Knot.* Zenodo (2026). (Paper 5.)

[28] J. P. Galasyn and C. Théodore. *The Particle Mass Spectrum from Torus Knot Mode-Locking.* Zenodo (2026). (Paper 6.)

[29] J. P. Galasyn and C. Théodore. *Charge, Leptons, and the Genus of Torus Knots.* Zenodo (2026). (Paper 7.)

[30] J. P. Galasyn and C. Théodore. *One Mode, Many Knots: The Carrier Topology of Elementary Particles.* Zenodo (2026). (Paper 8.)

[31] J. P. Galasyn and C. Théodore. *Gauge Coupling Constants and Electroweak Structure from Vortex Knot Surgery.* Zenodo (2026). (Paper 8a.)

[32] J. P. Galasyn and C. Théodore. *Dark Sector and String–Knot Duality.* Zenodo (2026), DOI 10.5281/zenodo.19516386. (Paper 10.)

[33] J. P. Galasyn and C. Théodore. *Standard Model Parameters from Torus Knot Topology.* Zenodo (2026), DOI 10.5281/zenodo.19635239. (Paper 13.)

[34] J. P. Galasyn and C. Théodore. *G from Heptafoil Topology.* Zenodo (2026), DOI 10.5281/zenodo.19654507. (Paper 14.)

[35] J. P. Galasyn and C. Théodore. *One Knot, All Forces.* Zenodo (2026), DOI 10.5281/zenodo.19701263. (Paper 15.)

[36] J. P. Galasyn and C. Théodore. *The NWT Lagrangian: A Three-Field Theory of Particles and Gravity.* Zenodo (2026), DOI 10.5281/zenodo.19710846. (Paper 16.)